PRINCIPLES OF ENGINEERING PHYSICS

1

Md. N. Khan • S. Panigrahi

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Principles of Engineering Physics 1

This is a textbook for an introductory course in engineering physics. It provides a coherent treatment of the basic principles and theories of engineering physics and offers a balance between theoretical concepts and their applications. Beginning with a comprehensive discussion on oscillations and waves with applications in the field of mechanical and electrical engineering, it goes on to explain basic concepts such as Huygen's principle, Fresnel's biprism, Fraunhofer diffraction and polarization.

All chapters are interspersed with rich pedagogical features such as solved problems, unsolved exercises and multiple choice questions with answers. It will help undergraduate students of engineering acquire skills for solving difficult problems in quantum mechanics, electromagnetism, nanoscience, energy systems and other engineering disciplines.

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To all our beloved people who have sacrificed their lives for the betterment of the world through science, technology and social service.

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Preface

Science in general may be described as organized common sense. In the real world of science, nothing prevails except rationality and logics. Science does not believe in miracles. Clear understanding of the basic principles of science is essential for technological and social development. Once upon a time, the base of engineering was mainly empirical; however, now it is completely scientific. Physics is a fundamental aspect of science on which all engineering sciences have been built upon. Nowadays, more stress is given to the understanding of the basic principles rather than on remembering specific procedures. The fundamental concepts of physics have paved the way for the development of technologies. All modern technological advances from laser micro surgery to television, from computers to dishwashers to mobile phones, from remote controlled toys to space vehicles, trace back directly to the principles of physics. Accordingly, the syllabus of engineering courses includes physics as an essential ingredient.

This book, entitled *Principles of Engineering Physics 1,* is designed as a textbook keeping in view the engineering physics course curricula prescribed by most technical universities of India. The present book begins with oscillations and waves and ends with holography, containing altogether fourteen chapters. This book is written in a logical and coherent manner for easy understanding. The concepts of physics are mathematized without losing the beauty of the physical ideas involved. Emphasis has been given to an understanding of the basic concepts and their applications to a number of engineering problems. Each topic has been discussed in detail, both conceptually and mathematically, so that students do not face any kind of difficulties. All the derivations and solutions of numerical examples are given in detail. Each chapter contains a large number of solved numerical examples, unsolved numerical problems with answers, practical applications, theoretical questions, and multiple choice questions with answers. Certain topics and derivations that are not included directly in the syllabi have also been included in the book for the sake of continuity and completeness. The scope of the book thus has been expanded beyond the basic needs of undergraduate engineering students. We hope this book will be of immense help not only to the students but also to the teachers.

The authors sincerely request the readers for their constructive criticisms via emails *mdnkhan1964@yahoo.com* and *spanigrahi@nitrkl.ac.in* for future modification of the book.

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It is a pleasure to express our deep appreciation to the engineering students (both continuing and passed out) of IGIT Sarang and NIT Rourkela who have borne with us in our class teachings. Many suggestions from our colleagues, students and reviewers have gone a long way in the development of this book. Our sincere thanks are due to them. We gratefully acknowledge the ideas received from a number of standard books on physics as given in the bibliography. We sincerely thank the editorial team at Cambridge University Press, India, for their keen interest in publishing this book in a nice format. We particularly wish to thank Gauravjeet Singh Reen for many helpful suggestions and improvements.

1 Oscillations and Waves

1.1 Introduction

Objects subjected to restoring forces when displaced from their normal positions and released, perform to and from or vibrating motions. They move back and forth along a path, repeating over and over again, a series of motions. Such motion of constant frequency is called periodic motion or harmonic motion and objects performing such type of motion are called harmonic oscillators. In this book, it is tacitly assumed that there is a linear relationship between force and displacement; frequency remains constant throughout the motion. In real systems however, the linear behavior, implicit in simple harmonic motion, is rarely obeyed. If the frequency of the oscillatory system is not constant, then it is called anharmonic motion – its study is beyond the scope of this book due to its mathematical complexities.

In oscillatory systems, it is not necessarily the bodies themselves who execute oscillations; bodies may be at rest. If the physical properties of a system undergo changes in an oscillatory manner, the system will also be called an oscillatory system. The electromagnetic energy transfer between the capacitor and inductor in a tank circuit used in electronic gadgets, variation of pressure in air due to propagation of sound waves, vibration of the diaphragm of a speaker in sound systems, flow of alternating current, variation of electric and magnetic vectors during propagation of electromagnetic waves, etc., are examples of oscillatory systems.

1.1.1 Parameters of an oscillatory system

i. *Mean position* The position of the oscillating body when there is no oscillation is called the mean position or equilibrium position. This is the rest position of the oscillating body.

ii. *Amplitude* (r) It is the absolute value of the maximum displacement of the oscillating particle from its mean position or equilibrium position.

iii. *Time period* (T) It is the time required for one complete oscillation.

iv. *Frequency* (ν) It is the number of complete oscillations made by the oscillating body in one second. The relation between frequency 'ν' and time period 'T' from definition is $T = \frac{1}{\nu}$.

1.2 Simple Harmonic Oscillation (SHO)

Let a body of mass '*m*' be placed on a frictionless plane with a massless spring attached to it (Fig. 1.1). The other end of the spring is fixed to a rigid support. The spring–body system is in the relaxed state, i.e., the spring is neither compressed nor extended. Notice the position of the body – it is called the mean position or equilibrium position. Now, the body is pulled through a displacement '*x*'. The spring exerts a restoring force on the body tending to pull it backwards. This restoring force '*F*' is proportional to the displacement (i.e., elongation of the spring) '*x*' and is opposite in direction to the displacement.

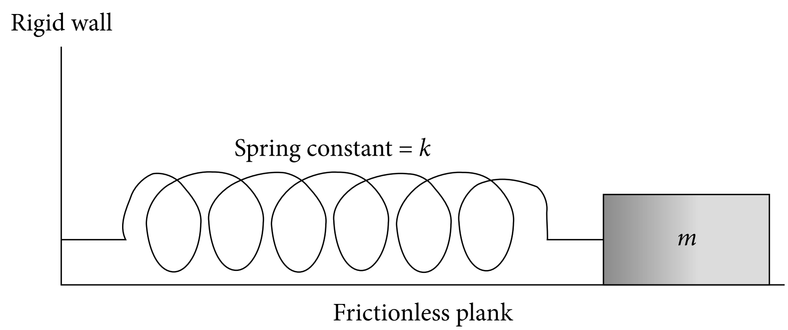

Figure 1.1 | Simple harmonic oscillator. Spring–body system is placed on a frictionless plane

Mathematically, $F \propto -x$ (negative sign appears since the restoring force and displacement '*x*' are in opposite directions)

$$F = -kx \tag{1.1}$$

where k is the proportionality constant and is known as spring constant. This equation is called Hooke's law of elasticity. Applying Newton's laws of motion, Hooke's law can be written as

$$ma = -kx$$

or $$m\frac{d^2x}{dt^2} = -kx \tag{1.2}$$

Equation (1.2) is the differential equation of motion of a simple harmonic oscillator in the absence of other forces. It can also be written as

$$\frac{d^2x}{dt^2}+\omega_0^2 x=0 \tag{1.3}$$

where $\omega_0^2=\dfrac{k}{m}$ (1.4)

Here m is the mass of the body attached at the end of the free end of the spring and ω_0 is called the natural angular frequency of oscillation. The general solution of the differential Eq. (1.3) is determined in the following way. Other methods for the purpose are also available.

Assume a solution of the form

$$x = e^{pt} \tag{1.5}$$

Putting (1.5) into (1.3), we get

$$p^2e^{pt}+\omega_0^2e^{pt}=0$$

or $\quad p=\pm i\omega_0$

The general solution of the differential Eq. (1.3) will be given by

$$x=C_1e^{i\omega_0 t}+C_2e^{-i\omega_0 t} \tag{1.6}$$

The constants C_1 and C_2 must be complex in order for Eq. (1.6) to be the general solution. Since $e^{i\omega_0 t}$ and $e^{-i\omega_0 t}$ are complex conjugates of each other, the constants C_1 and C_2 must be complex conjugate of each other so that x is real. For this, we set

$$C_1=C=Ae^{i\theta}$$

and $\quad C_2=C^*=Ae^{-i\theta}$

Upon this substitution into Eq. (1.6), we get

$$x=Ce^{i\omega_0 t}+C^*e^{-i\omega_0 t}=Ae^{i\theta}e^{i\omega_0 t}+Ae^{-i\theta}e^{-i\omega_0 t}$$

or $\quad x=2A\cos(\omega_0 t+\theta)=r\cos(\omega_0 t+\theta),\ 2A=r$ (1.7)

($x=r\sin(\omega_0 t+\theta)$ may also be a solution).

Here

r = amplitude of the oscillation.

θ = the initial phase of the oscillation.

$\omega_0 t + \theta$ = phase of the motion.

The constants r and θ can be obtained from the initial conditions of the simple harmonic motion.

The velocity 'v' of the body executing SHO is determined by differentiating Eq. (1.7) with respect to time.

$$v = -r\omega_0 \sin(\omega_0 t + \theta) \tag{1.8}$$

The acceleration 'a' of the body executing SHO is determined by differentiating Eq. (1.8) with respect to time.

$$a = -\omega_0^2 r \cos(\omega_0 t + \theta)\,. \tag{1.9}$$

Putting the value of $r\cos(\omega_0 t + \theta) = x$ from (1.7) into Eq. (1.9), we get

$$a = -\omega_0^2 x \tag{1.10}$$

Equation (1.10) shows that in the case of a simple harmonic motion, acceleration is directed opposite to displacement.

If T is the time period, then in T seconds, the number of complete oscillations is 1. So, in 1 second, the number of oscillations will be $1/T$ which by definition is the frequency ν. Hence, we have

$$\nu = \frac{1}{T} \tag{1.11}$$

Frequency and time period are inversely proportional to each other.

From the definition of angular frequency, we have

$$\omega_0 = \frac{2\pi}{T} = 2\pi\nu \tag{1.12}$$

1.2.1 Energy of a simple harmonic oscillator

By the definition of potential energy, the potential energy of a simple harmonic oscillator at any instant or at any position is given by

$$E_P = \int_0^x F dx = \int_0^x kx dx = \frac{1}{2}kx^2 \tag{1.13}$$

Here 'x' has been measured from the mean position $x = 0$. Putting the value of x from Eq. (1.7) into Eq. (1.13), we have

$$E_P = \frac{1}{2}kr^2 \cos^2(\omega_0 t + \theta) \tag{1.14}$$

Putting the value of k from Eq. (1.4) into Eq. (1.14), we get

$$E_P = \frac{1}{2}m\omega_0^2 r^2 \cos^2(\omega_0 t + \theta) \tag{1.15}$$

Since the maximum value of $\cos^2(\omega_0 t + \theta)$ is 1, the maximum value of potential energy-$E_{P\max}$ from Eq. (1.15) is found out to be

$$E_{P\max} = \frac{1}{2}m\omega_0^2 r^2 \tag{1.16}$$

By the definition of kinetic energy, the kinetic energy of a simple harmonic oscillator at any instant or at any position is given by

$$E_K = \frac{1}{2}mv^2$$

Putting the value of v from Eq. (1.8) into the previous equation, we get

$$E_K = \frac{1}{2}mr^2\omega_0^2 \sin^2(\omega_0 t + \theta) \tag{1.17}$$

Since the maximum value of $\sin^2(\omega_0 t + \theta)$ is 1, the maximum value of kinetic energy $E_{K\max}$ from Eq. (1.17) is found out to be

$$E_{K\max} = \frac{1}{2}m\omega_0^2 r^2 \tag{1.18}$$

The total energy of a simple harmonic oscillator at any instant or at any position is given by

$$E = E_P + E_K$$

$$= \frac{1}{2}m\omega_0^2 r^2 \cos^2(\omega_0 t + \theta) + \frac{1}{2}mr^2\omega_0^2 \sin^2(\omega_0 t + \theta)$$

$$= \frac{1}{2}m\omega_0^2 r^2 = E_{K\max} = E_{P\max} \tag{1.19}$$

Equation (1.19) shows that the total energy of a simple harmonic oscillator at any instant or at any position is constant.

1.2.2 Characteristics of SHO

From the discussions in the previous section, we can infer the following characteristics of SHO.

When the body is released, it moves to and fro about the mean position with constant amplitude 'r'. As the body moves towards the mean position, its speed increases but the force and hence, its acceleration decreases; both become zero at the mean position. Due to the inertia of motion, the body overshoots the mean position, but at the same time, a retarding force comes into action to oppose the motion. This retarding force increases until the body reaches the largest distance from the mean position. Here, it stops and begins its return journey. This process goes on continuously for an indefinite period if there is no dissipative force. Throughout the motion, the force as well as the acceleration is directly proportional to displacement and directed towards the mean position. The distances travelled by the vibrating body on the two sides of the mean position are equal. This type of motion is called simple harmonic oscillation. The SHO is graphically represented in Fig. 1.2.

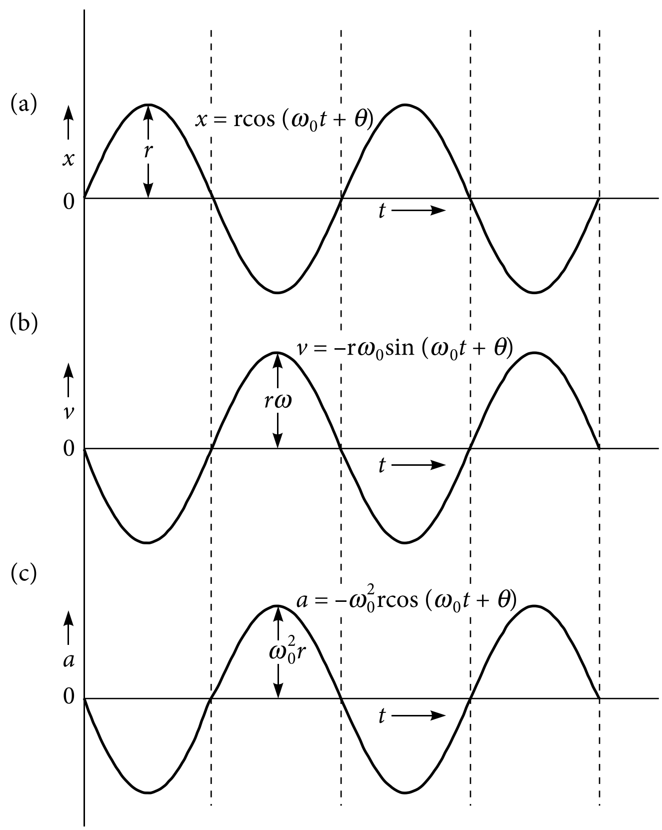

Figure 1.2 Graphical representation of SHO. (a) displacement-time plot, (b) velocity–time plot, (c) acceleration–time plot

For a particular simple harmonic oscillator, mass, amplitude and angular frequency are constants with respect to time. Therefore, we can conclude that the total energy of a simple harmonic oscillator at any instant or position is constant. Moreover, the total energy of a simple harmonic oscillator at any instant or position is equal to the maximum values of potential or kinetic energy. The simultaneous variation of potential energy and kinetic energy of a simple harmonic oscillator according to Eqs (1.15) and (1.17) is depicted in Fig. 1.3(a) and in Fig. 1.3(b).

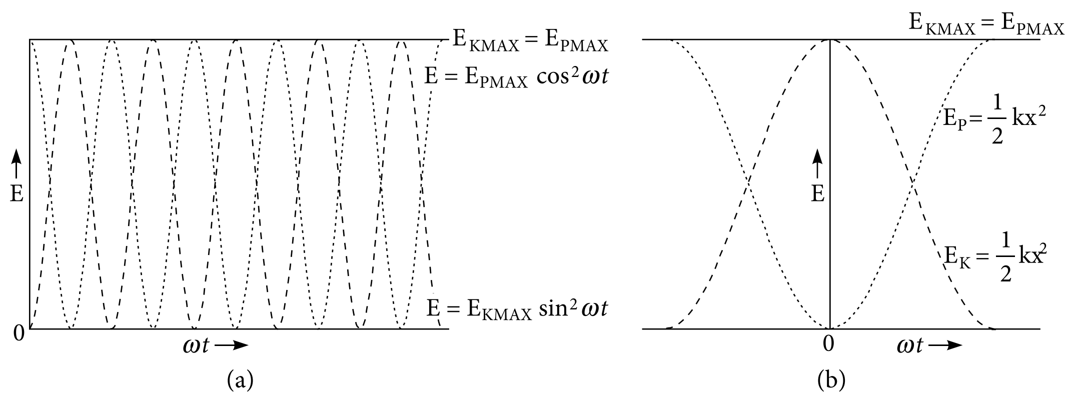

Figure 1.3 Energy of a simple harmonic oscillator. The dotted curve represents potential energy, the dashed line curve represents kinetic energy and the continuous curve represents the total energy of a simple harmonic oscillator

We will now cite few examples of simple harmonic motion under ideal conditions. The derivations for time periods in all the examples are left as exercises to the students.

i. The motion of a simple pendulum in vacuum is simple harmonic. Its time period is given by

$$T = 2\pi\sqrt{\frac{\ell}{g}},\quad \ell = \text{length of the simple pendulum.}$$

ii. Take a cleaned U-shaped glass tube and fix it vertically on a stand. Partially fill it with mercury. Now the levels of the mercury column in both sides are equal. Blow slowly through one end so that the mercury column in the other end rises slightly. When we stop blowing, the mercury column executes simple harmonic motion having time period

$$T = 2\pi\sqrt{\frac{\ell}{2g}},\quad \ell = \text{length of the mercury column in the U-tube.}$$

iii. A body dropped into a hole dug through the centre of earth executes simple harmonic motion with time period

$$T = 2\pi\sqrt{\frac{R}{g}},\ R = \text{radius of Earth.}$$

iv. Place a small block inside a smooth curved surface having radius of curvature 'R'. Move the sphere slightly along the curved surface away from the equilibrium position and then release. It will perform simple harmonic motion with time period

$$T = 2\pi\sqrt{\frac{R}{g}}.$$

v. Place a sphere of radius 'r' inside a curved surface of radius of curvature 'R'. Move the sphere slightly along the curved surface away from the equilibrium position and then release. It will execute simple harmonic motion with time period

$$T = 2\pi\sqrt{\frac{7(R-r)}{5g}} \approx 2\pi\sqrt{\frac{7R}{5g}} \quad \text{if } R >> r$$

vi. If a small iron cylinder, partially submerged in mercury vertically, is pressed slightly and then released, it will execute simple harmonic motion with time period

$T = 2\pi\sqrt{\frac{\ell\rho}{\rho' g}}$. Here ℓ = length of the cylinder, ρ = density of the cylinder, ρ' = density of mercury.

Example 1.1

A spring 15 cm in length is fixed to the ceiling. When a body of mass 1 kg is hung at the free end, its length becomes 17 cm. Calculate the spring constant of the spring.

Solution

The increase in length 'x' of the spring when a 1kg body is hung = 17 cm – 15 cm = 2 cm = 0.02 m

The downward force acting on the spring = weight of the body hung. At the equilibrium position, the restoring force = downward force acting on the spring.

$$kx = mg$$

or $$k = \frac{mg}{x} = \frac{1 \times 9.8\ \text{N}}{0.02\ \text{m}} = 490\ \text{N/m}$$

The spring constant of the spring is calculated to be 490 N/m.

Example 1.2

A 5 kg body extends a spring 15 cm from its relaxed position. The body is removed and a 1 kg body is hung from the same spring. The 1 kg body is pulled and released. Calculate the time period of oscillation of the body.

Solution

The increase in length 'x' of the spring when a 5 kg body is hung = 15 cm = 0.15 m

The downward force acting on the spring = weight of the body hung. At the equilibrium position, the restoring force = downward force acting on the spring.

$$kx = mg$$

or $k = \dfrac{mg}{x} = \dfrac{5\times 9.8\ \text{N}}{0.15\ \text{m}} = 326.67\ \text{N/m}$ is the spring constant of the spring.

The natural angular frequency of the oscillation $\omega_0 = \sqrt{\dfrac{k}{m}}$.

The time period of oscillation $T = 2\pi\sqrt{\dfrac{m}{k}} = 2\pi\sqrt{\dfrac{1}{326.67}}\ \text{s} = 0.35\ \text{s}$

Example 1.3

A body oscillates with simple harmonic motion obeying the equation $y = 12\ \cos\left(0.7\pi t + \dfrac{\pi}{5}\right)$ m.

Calculate the velocity and acceleration of the body at time t = 3 s. Also calculate the natural frequency and time period of the harmonic motion.

Solution

The velocity 'v' of the body at any time t is

$$v = \frac{dy}{dt} = \frac{d}{dt}\left[12\cos\left(0.7\pi t + \frac{\pi}{5}\right)\right]\text{m/s} = -8.4\pi\sin\left(0.7\pi \text{t} + \frac{\pi}{5}\right)\text{m/s}$$

So the velocity of the body at t = 3 s will be

$$v = -8.4\pi\sin\left(0.7\pi\times 3 + \frac{\pi}{5}\right)\ \text{m/s} = -21.35\ \text{m/s}$$

The velocity is directed opposite to the displacement.

The acceleration 'a' of the body at any time t is

$$a = \frac{dv}{dt} = \frac{d}{dt}\left[-8.4\pi\sin\left(0.7t\pi + \frac{\pi}{5}\right)\right]\text{m/s}^2 = -5.88\pi^2\cos\left(0.7\pi t + \frac{\pi}{5}\right)\text{m/s}^2$$

So the acceleration of the body at t = 3 s will be

$$a = -5.88\pi^2 \cos\left(0.7\pi \times 3 + \frac{\pi}{5}\right) \text{ m/s}^2 = -34.11 \text{ m/s}^2.$$

The acceleration is directed opposite to the displacement.

The initial phase constant $= \frac{\pi}{5}$.

The angular frequency $\omega = 0.7\pi \text{s}^{-1}$

Hence, the frequency of the oscillation $\nu = \frac{0.7\,\text{s}^{-1}}{2} = 0.35\,\text{s}^{-1}$

The time period of the oscillation $T = \frac{1}{\nu} = 2.86\,\text{s}$

1.3 Damped Harmonic Oscillation (DHO)

In case of simple harmonic oscillation, the amplitude of oscillation does not decrease. However in reality, in the case of simple pendulums, amplitude decreases with the passing of time. This is due to the viscosity of the medium. The force is the dissipative force F_d. Harmonic oscillation under the influence of a spring–like restoring force in a viscous medium is called damped harmonic oscillation.

For a low velocity dissipative force, F_d is directly proportional to the velocity of the oscillating body and is always directed opposite to the velocity of the body. Mathematically, we have

$$F_d = -b\frac{dx}{dt} \tag{1.20}$$

The minus sign appears because the direction of the dissipative force F_d is opposite to that of velocity. The restoring force on the oscillating body from Eq. (1.1) is $-kx$. Therefore, the total force on the vibrating body in a dissipative medium is

$$F_{\text{total}} = -b\frac{dx}{dt} - kx \tag{1.21}$$

or $$ma = -b\frac{dx}{dt} - kx$$

or $$\frac{d^2x}{dt^2} + \frac{b}{m}\frac{dx}{dt} + \frac{k}{m}x = 0 \tag{1.22}$$

Let $\frac{b}{m} = 2\gamma$ and $\frac{k}{m} = \omega_0^2$

γ and $\frac{\omega_0}{2\pi}$ are the damping coefficient of the dissipative medium and the natural frequency of the damped harmonic oscillator. Putting these substitutions into Eq. (1.22), we get

$$\frac{d^2x}{dt^2} + 2\gamma\frac{dx}{dt} + \omega_0^2 x = 0 \tag{1.23}$$

Equation (1.23) is the differential equation of motion of a damped harmonic oscillation in a dissipative medium. The solutions of this differential equation depend upon the relative values of γ and ω_0. The general solution of the differential Eq. (1.23) is determined in the following way. Other methods for the purpose are also available.

Assuming a solution of the form $x = e^{pt}$, from Eq. (1.23), we have

$$p^2 + 2\gamma p + \omega_0^2 = 0$$

or
$$p = -\gamma \pm \sqrt{\gamma^2 - \omega_0^2} \tag{1.24}$$

Depending upon the values of γ, three different situations can arise for a damped harmonic oscillator. If the medium and frequency of the oscillator is such that the $\gamma < \omega_0$ condition is satisfied, the oscillator is said to be underdamped. If the medium and frequency of the oscillator is such that the $\gamma > \omega_0$ condition is satisfied, the oscillator is said to be overdamped. If the medium and frequency of the oscillator is such that the $\gamma = \omega_0$ condition is satisfied, the oscillator is said to be critically damped.

Case 1: $\gamma < \omega_0$ (Under damped)

From Eq. (1.24), we have

$$p = -\gamma \pm i\omega_1 \quad \text{where} \quad \omega_1 = \sqrt{\omega_0^2 - \gamma^2}$$

The general solution of the differential Eq. (1.23) will be given by

$$x = C_1 e^{-\gamma t + i\omega_1 t} + C_2 e^{-\gamma t - i\omega_1 t} \tag{1.25}$$

The constants C_1 and C_2 must be complex in order that Eq. (1.25) is the general solution. Since $e^{-\gamma t + i\omega_1 t}$ and $e^{-\gamma t - i\omega_1 t}$ are complex conjugates of each other, the constants C_1 and C_2 must be complex conjugates of each other, so that x is real. For this, we set

$$C_1 = C = Ae^{i\theta}$$

and $C_2 = C^* = Ae^{-i\theta}$

Upon this substitution in Eq. (1.25), we get

$$x = Ce^{-\gamma t + i\omega_1 t} + C^* e^{-\gamma t - i\omega_1 t} = Ae^{i\theta} e^{-\gamma t + i\omega_1 t} + Ae^{-i\theta} e^{-\gamma t - i\omega_1 t}$$

or $$x = Ae^{-\gamma t}\left(e^{i(\omega_1 t + \theta)} + e^{-i(\omega_1 t + \theta)}\right)$$

or $$x = 2Ae^{-\gamma t}\cos\left(\omega_1 t + \theta\right)$$

Putting $2A = r$ into the aforementioned equation, we get

$$x = re^{-\gamma t}\cos\left(\omega_1 t + \theta\right) \tag{1.26}$$

Here the values of the constants 'r' and 'θ' depend upon the initial conditions. The frequency of oscillation in this case is $\frac{\omega_1}{2\pi}$ and the amplitude is $re^{-\gamma t}$ which decreases exponentially with time. The phenomenon is plotted in the Fig. 1.4. This frequency does not change as the oscillations decay.

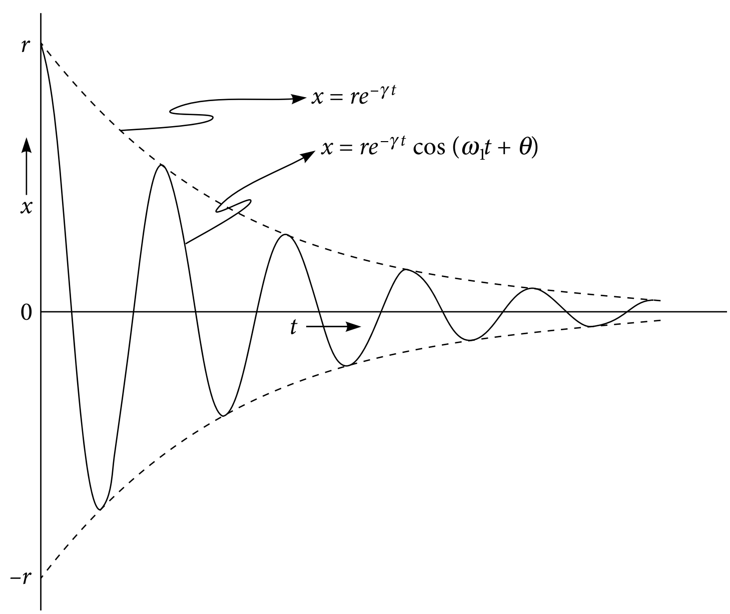

Figure 1.4 Motion of a damped harmonic oscillator. The solid curve is the plot of $x = re^{-\gamma t}\cos\omega t$ and the dashed curve is the plot of $x = \pm re^{-\gamma t}$

The total energy of the damped harmonic oscillator according to Eq. (1.19) may be given as

$$E = E_{P\max} = \left(\frac{1}{2}kx^2\right)_{\max} = \left(\frac{1}{2}kr^2e^{-2\gamma t}\cos^2(\omega_1 t+\theta)\right)_{\max}$$

$$= \frac{1}{2}m\omega_0^2 r^2 e^{-2\gamma t}$$

or $$E = E_0 e^{-2\gamma t} \tag{1.27}$$

Here $E_0 = \frac{1}{2}m\omega_0^2 r^2$ is the total initial energy of the damped harmonic oscillator at t = 0. Thus, the total energy decreases more rapidly than the amplitude with time.

Case 2: $\gamma > \omega_0$ (Over damped)

If $\gamma > \omega_0$, $\sqrt{\gamma^2 - \omega_0^2}$ is real and the two solutions of p are

$$p = -\gamma - \sqrt{\gamma^2 - \omega_0^2} = -\gamma_1$$

and $$p = -\gamma + \sqrt{\gamma^2 - \omega_0^2} = -\gamma_2$$

The general solution of Eq. (1.23) will be obtained as

$$x = C_1 e^{-\gamma_1 t} + C_2 e^{-\gamma_2 t} \tag{1.28}$$

In this case, both terms decrease exponentially with time, one at a faster rate than the other. In the overdamped condition, the body does not oscillate. The values of the constants C_1 and C_2 depends upon the initial conditions.

Case 3: $\gamma = \omega_0$ (Critically damped)

If $\gamma = \omega_0$, Eq. (1.23) has two solutions namely $x = e^{-\gamma t}$ and $x = te^{-\gamma t}$. Therefore, in this case of critically damped oscillations, the general solution of Eq. (1.23) will be

$$x = (C_1 + C_2 t)e^{-\gamma t} \tag{1.29}$$

and declines exponentially at a faster rate than that of Case 2. In this case, the body comes to rest in finite time without oscillation.

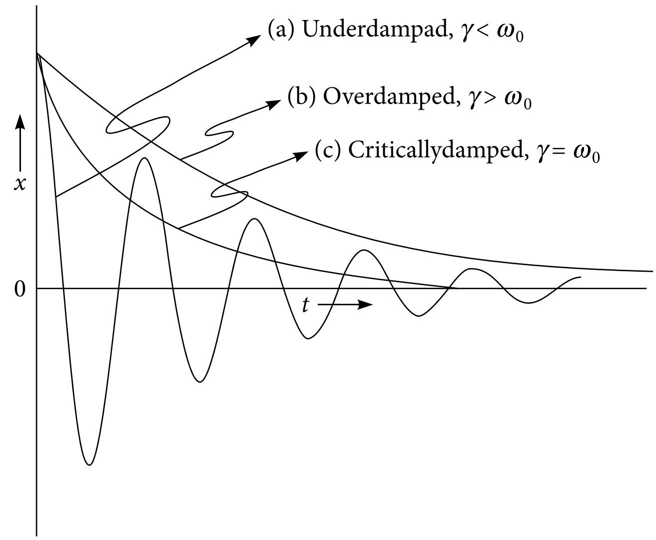

Figure 1.5 Return of a harmonic oscillator to equilibrium position. (a) Under damped (b) Over damped (c) Critically damped

All the three cases are plotted in Fig. 1.5.

The pointer reading meters, hydraulic and pneumatic spring returns for door are so designed that they come to equilibrium/rest position quickly without over shoots or oscillation. These devices are so designed that the condition of critical damping $\gamma = \omega_0$ is satisfied. If the condition of over damping $\gamma > \omega_0$ is satisfied in these devices, it will take a long time to come to equilibrium position. If the condition of under–damping $\gamma < \omega_0$ is satisfied in these devices, it will over shoot the equilibrium position creating noise and will thus affect the life span of the devices.

1.3.1 Damping of an oscillator

Damping is an effect that reduces the amplitude of oscillations in an oscillatory system, particularly the harmonic oscillator. In a simple harmonic oscillator, total energy $\left(E_0 = \frac{1}{2} m\omega_0^2 r^2\right)$ which is proportional to square of amplitude remains constant throughout time and positions. Due to the presence of damping force $-2m\gamma \frac{dx}{dt}$ in the case of under–damped harmonic motion, the amplitude R and energy E of the oscillator becomes $R = re^{-\gamma t}$ and $E = E_0 e^{-2\gamma t}$ respectively. The larger the value of the damping coefficient γ, the more rapidly does the amplitude and energy decrease. Measuring natural decay in terms of the fraction e^{-1} of the original value is a very common practice in physics. Thus, we can make use of the exponential factor to express the rates at which the amplitude and energy are decreased. The time for a natural decay process to reach zero is of course theoretically infinite. We shall discuss the damping of an oscillator in three different ways.

1.3.1(a) Logarithmic decrement

Logarithmic decrement λ measures the rate of decrease of amplitude of an under–damped harmonic motion. The under–damped harmonic motion is described by the equation $x = re^{-\gamma t}\cos(\omega_1 t + \theta)$, where amplitude is $R = re^{-\gamma t}$. The equation shows that amplitude decreases exponentially with time. Time period $\frac{2\pi}{\omega_1}$ remains constant. Let T be the time period of the under–damped harmonic motion. Let R_0, R_1, R_2, R_3, and R_4 be the amplitudes at time 0, T, $2T$, $3T$, and $4T$ respectively. Then, we have

$$R_O = re^{-\gamma 0} = r,\ R_1 = re^{-\gamma T},\ R_2 = re^{-\gamma 2T},\ R_3 = re^{-\gamma 3T},\ R_4 = re^{-\gamma 4T}, \text{ etc.}$$

Now we can have $\frac{R_0}{R_1} = \frac{R_1}{R_2} = \frac{R_2}{R_3} = \frac{R_3}{R_4} = \frac{R_4}{R_5} = e^{\gamma T} = e^{\lambda}$

where $\lambda = \gamma T$ is called logarithmic decrement. Taking the natural logarithm of the aforementioned equation, we get

$$\ln\frac{R_0}{R_1} = \ln\frac{R_1}{R_2} = \ln\frac{R_2}{R_3} = \ln\frac{R_3}{R_4} = \ln\frac{R_4}{R_5} = \ln e^{\gamma T} = \ln e^{\lambda} = \lambda$$

In general, we can have

$$\ln\frac{R_{n-1}}{R_n} = \lambda = \gamma T \tag{1.30}$$

Now we can define the logarithmic decrement in the following way. Logarithmic decrement is defined as the natural logarithm of the ratio of two consecutive and successive amplitudes which are separated by one time period. In Eq. (1.30), amplitudes R_n and R_{n-1} are separated by one time period.

Example 1.4

A point performs a damped harmonic oscillation with frequency ω_1 and damping coefficient γ. Find the initial amplitude 'r' and the initial phase θ if at the moment $t = 0$ the displacement of the point and its velocity are $y(0) = 0$ and $v(0) = b$ respectively.

Solution

The equation of an oscillatory damped harmonic motion is given by

$$y(t) = re^{-\gamma t}\cos(\omega_1 t + \theta) \tag{A}$$

The velocity of the oscillating point is given by

$$v(t) = \frac{dy}{dt} = r\omega_1 e^{-\gamma t} \sin(\omega_1 t + \theta) - r\gamma e^{-\gamma t} \cos(\omega_1 t + \theta) \tag{B}$$

Putting $t = 0$ into (A), we get

$$y(0) = re^{-\gamma 0} \cos\left(\omega_1 \times 0 + \theta\right) = r\cos\theta$$

or $\quad 0 = r\cos\theta$

or $\quad \theta = \frac{\pi}{2}$ or $-\frac{\pi}{2}$

Case 1: Suppose $\theta = \frac{\pi}{2}$

Putting $t = 0$ and $\theta = \frac{\pi}{2}$ into (B), we get

$$v(0) = r\omega_1\, e^{-\gamma 0} \sin\left(\omega_1 \times 0 + \frac{\pi}{2}\right) - r\gamma e^{-\gamma 0} \cos\left(\omega_1 \times 0 + \frac{\pi}{2}\right)$$

$$b = r\omega_1 \times 1$$

$$r = \frac{b}{\omega_1} \text{ when } \theta = \frac{\pi}{2}$$

Case 2: Suppose $\theta = -\frac{\pi}{2}$

Putting $t = 0$ and $\theta = -\frac{\pi}{2}$ into (B), we get

$$v(0) = r\omega_1\, e^{-\gamma 0} \sin\left(\omega_1 \times 0 - \frac{\pi}{2}\right) - r\gamma\, e^{-\gamma 0} \cos\left(\omega_1 \times 0 - \frac{\pi}{2}\right)$$

$$b = r\omega_1 \times (-1)$$

$$r = \frac{-b}{\omega_1} \text{ when } \theta = -\frac{\pi}{2}$$

Example 1.5

The amplitude of an under–damped oscillator falls to 1/10 of its initial value after 10 oscillations. If the time period is 2 s, calculate the (a) damping coefficient, (b) logarithmic decrement, (c) time during which energy falls to 1/10 of its original value.

Solution

The amplitude of the under–damped harmonic oscillation after n number of oscillations is given by

$$R_n = re^{-\gamma nT}$$

where T is the time period.

or $$\frac{r}{R_n} = e^{\gamma nT}$$

(a) According to the problem, $n = 10$, $T = 2$ s, $\frac{r}{R_{10}} = 10$

So we have

$$10 = e^{\gamma \times 10 \times 2}$$

or $$20\gamma = \ln 10$$

or $$\gamma = \frac{2.3}{20} = 0.115$$

(b) Logarithmic decrement λ is given by

$$\lambda = \gamma T = 0.115 \times 2 = 0.23$$

(c) The energy of an under–damped harmonic oscillator is given by

$$E = E_0 e^{-2\gamma t}$$

or $$\frac{E}{E_O} = e^{-2\gamma t}$$

According to the given data $\frac{E}{E_0} = \frac{1}{10}$

or $\frac{1}{10} = e^{-2\gamma t}$

or $e^{2\gamma t} = 10$

or $2\gamma t = \ln 10$

or $t = \frac{\ln 10}{2 \times 0.115} = 10\text{s}.$

Example 1.6

A body performs damped harmonic oscillation according to the equation $x = re^{-\gamma t} \sin \omega t$. Calculate the (a) amplitude of oscillation and (b) velocity of the body at time t = 0; (c) the moments of time at which body reaches the extreme positions.

Solution

(a) The equation of motion of the damped harmonic oscillator is given as $x = re^{-\gamma t} \sin \omega t$. The amplitude of oscillation at any time is

$$R_t = re^{-\gamma t}.$$

So the amplitude of oscillation at time t = 0 will be

$$R_O = re^{-\gamma \times 0} = r$$

(b) The instantaneous velocity v of the damped harmonic oscillator at any time is given by

$$v = \frac{d}{dt} re^{-\gamma t} \sin \omega t = -r\gamma e^{-\gamma t} \sin \omega t + r\omega e^{-\gamma t} \cos \omega t \tag{A}$$

So the velocity v of the damped harmonic oscillator at time t = 0 will be

$$v(0) = r\omega$$

Since the body is performing damped harmonic motion, the velocity of the body at extreme positions is zero. If the body reaches the extreme positions at time t_n, then from (A), we have

$$r\gamma e^{-\gamma t_n} \sin \omega t_n = r\omega e^{-\gamma t_n} \cos \omega t_n$$

or $t_n = \frac{1}{\omega} \left(\tan^{-1} \frac{\omega}{\gamma} + n\pi \right)$

Example 1.7

A body performs damped harmonic oscillation according to the equation $x = re^{-\gamma t}\cos(\omega t + \theta)$. Calculate the (a) initial amplitude r of oscillation and (b) initial phase θ if at time $t = 0$, velocity of the body is u and $x = 0$.

Solution

According to the problem, at $t = 0$, $x = 0$. So putting $t = 0$ and $x = 0$ in the equation $x = re^{-\gamma t}\cos(\omega t + \theta)$, we get

$$0 = re^{-\gamma 0}\cos\left(\omega \times 0 + \theta\right)$$

or $\cos\theta = 0$

or $\theta = \pm\dfrac{\pi}{2}$

The instantaneous velocity v of the damped harmonic oscillator at any time is given by

$$v = \frac{d}{dt} re^{-\gamma t}\cos(\omega t + \theta)$$

$$= -r\gamma e^{-\gamma t}\cos\left(\omega t + \theta\right) - r\omega e^{-\gamma t}\sin\left(\omega t + \theta\right) \quad \text{(A)}$$

According to the problem, at $t = 0$, $v = u$. So putting $t = 0$ and $v = u$ in (A), we get

$$u = -r\gamma\cos\theta - r\omega\sin\theta \quad \text{(B)}$$

If $\theta = \dfrac{\pi}{2}$ is put in (B), u becomes negative and if $\theta = -\dfrac{\pi}{2}$ is put in (B), u becomes positive. Thus, we have

$\theta = \dfrac{\pi}{2}$ if $u < 0$

and

$\theta = -\dfrac{\pi}{2}$ if $u > 0$

Putting $\theta = \pm\dfrac{\pi}{2}$ in (B), we get

$$|u| = r\omega$$

or Initial amplitude of oscillation r will be $r = \frac{|u|}{\omega}$

1.3.1(b) *Modulus of decay*

The amplitude of an under–damped harmonic oscillator is given as

$$R_t = R_0 e^{-\gamma t}$$

The time interval during which amplitude decreases to $\frac{1}{e}$ of the original amplitude is called modulus of decay or relaxation time τ. At $t = \frac{1}{\gamma}$ seconds, the amplitude becomes $\frac{R_0}{e}$ and hence, $\frac{1}{\gamma}$ is the modulus of decay or relaxation time i.e.,

$$\tau = \frac{1}{\gamma} \tag{1.31}$$

1.3.1(c) *Quality factor of an under damped harmonic oscillator*

This measures the rate at which the energy decays. The energy of an under damped harmonic oscillator at any instant of time is given as

$$E = E_0 e^{-2\gamma t} \tag{1.32}$$

The quality factor Q is defined as the number of radians through which an under–damped system goes so that its energy decreases to $\frac{E_0}{e}$. During time interval $\frac{1}{2\gamma}$, energy decreases to $\frac{E_0}{e}$.

During time interval T seconds (time period), the oscillator traces 2π radians. Then during the time interval $\frac{1}{2\gamma}$ seconds, the oscillator traces $\frac{2\pi}{T} \times \frac{1}{2\gamma} = \frac{\omega_1}{2\gamma}$ radians. By definition, quality factor Q is given by

$$Q = \frac{\omega_1}{2\gamma} = \frac{1}{2}\omega_1 \tau \tag{1.33}$$

In the aforementioned expression, ω_1 is the angular frequency of the under–damped oscillator.

From Eq. (1.32), we get

$$-dE = 2\gamma E_0 e^{-2\gamma t} dt$$

So the energy lost per cycle

$$-dE = 2\gamma E_0 e^{-2\gamma t} \times T = E_0 e^{-2\gamma t} \times \frac{2\pi}{Q}$$

or $$\frac{-dE}{E_0 e^{-2\gamma t}} = \frac{2\pi}{Q}$$

or $$Q = 2\pi \frac{E_0 e^{-2\gamma t}}{-dE}$$

Thus, quality factor can be defined as

$$Q = 2\pi \frac{\text{Energy stored in the system}}{\text{Energy lost per cycle}} \tag{1.34}$$

1.4 Forced Vibrations

The real life phenomena of forced vibrations can be observed in the vibration of a bridge under the influence of marching soldiers, vibration of an electric motor due to the periodic impulses from an irregularity in the shaft, vibration of a tuning fork when subjected to the periodic force of a sound wave, and so on. The harmonic oscillations under the influence of an externally applied sinusoidally varying force are called forced vibrations.

Let the externally applied sinusoidally varying force be

$$F_0 \cos(\omega t + \theta_0) \tag{1.35}$$

where

F_0 = Amplitude of the externally applied sinusoidally varying force.

ω = Angular frequency of the externally applied sinusoidally varying force.

θ_0 = Initial phase of the externally applied sinusoidally varying force.

From Eq. (1.20), the dissipative force on the oscillator is $F_d = -b\frac{dx}{dt}$ and from Eq. (1.1), the restoring force on the oscillating body is $-kx$. Therefore, the total force on the vibrating body in this case is

$$F_{\text{Total}} = -b\frac{dx}{dt} - kx + F_0 \cos(\omega t + \theta_O)$$

By applying Newton's law of motion to the aforementioned equation, the equation of motion of a forced harmonic oscillator is obtained as

$$\frac{d^2x}{dt^2} + 2\gamma\frac{dx}{dt} + \omega_0^2 x = \frac{F_0}{m}\cos\left(\omega t + \theta_0\right) \tag{1.36}$$

where $\frac{b}{m} = 2\gamma$ and $\frac{k}{m} = \omega_0^2$. γ and $\frac{\omega_0}{2\pi}$ are the damping coefficients of the dissipative medium and the natural frequency of an undamped harmonic oscillator respectively. Equation (1.36) is the differential equation of motion of a forced vibration in a dissipative medium. The general solution of the differential Eq. (1.36) is determined in the following way using complex functions. Other methods for the purpose are also available. Equation (1.36) is modified to

$$\frac{d^2x}{dt^2} + 2\gamma\frac{dx}{dt} + \omega_0^2 x = \frac{1}{m}F_0 e^{i\theta_0} e^{i\omega t} \tag{1.37}$$

where $\frac{F_0}{m}\cos\left(\omega t + \theta_0\right)$ is the real part of $\frac{1}{m}F_0 e^{i\theta_0} e^{i\omega t}$, i.e.,

$$\mathrm{Re}\left(\frac{1}{m}F_0 e^{i\theta_0} e^{i\omega t}\right) = \frac{F_0}{m}\cos\left(\omega t + \theta_0\right)$$

Assume a complex solution of the differential Eq. (1.37) of the form

$$x = x_0 e^{i\omega t} \tag{1.38}$$

Hence, the complex velocity of the forced oscillator is

$$\frac{dx}{dt} = i\omega x_0 e^{i\omega t} \tag{1.39}$$

and the complex acceleration of the forced oscillator is

$$\frac{d^2x}{dt^2} = -\omega^2 x_0 e^{i\omega t} \tag{1.40}$$

Putting Eq. (1.38) into Eq. (1.40) and Eq. (1.37), we get

$$-\omega^2 x_0 e^{i\omega t} + i2\gamma\omega x_0 e^{i\omega t} + \omega_0^2 x_0 e^{i\omega t} = \frac{1}{m}F_0 e^{i\theta_0} e^{i\omega t}$$

or $$x_0 = \frac{\frac{F_0}{m} e^{i\theta_0}}{\omega_0^2 - \omega^2 + 2i\gamma\omega} = \frac{\frac{F_0}{m} e^{i\theta_O}}{\left[\left(\omega_0^2 - \omega^2\right) + 4\gamma^2\omega^2\right]^{\frac{1}{2}} e^{i\tan^{-1}\frac{2\gamma\omega}{\omega_0^2 - \omega^2}}} \tag{1.41}$$

Let

$$\tan^{-1}\frac{2\gamma\omega}{\omega_0^2 - \omega^2} = \frac{\pi}{2} - \beta \tag{1.42}$$

Upon this substitution in Eq. (1.41), we get

$$x_0 = \frac{\frac{F_0}{m} e^{i\theta_0}}{\left[\left(\omega_0^2 - \omega^2\right) + 4\gamma^2\omega^2\right]^{\frac{1}{2}} e^{i\left(\frac{\pi}{2} - \beta\right)}} = \frac{\frac{F_0}{m} e^{i\theta_0} e^{-i\frac{\pi}{2}} e^{i\beta}}{\left[\left(\omega_0^2 - \omega^2\right) + 4\gamma^2\omega^2\right]^{\frac{1}{2}}}$$

$$= \frac{-i\frac{F_0}{m} e^{i(\theta_0 + \beta)}}{\left[\left(\omega_0^2 - \omega^2\right) + 4\gamma^2\omega^2\right]^{\frac{1}{2}}}$$

or $$x_0 = \frac{\frac{F_0}{m} e^{i(\theta_0 + \beta)}}{i\left[\left(\omega_O^2 - \omega^2\right) + 4\gamma^2\omega^2\right]^{\frac{1}{2}}} \tag{1.43}$$

Putting this value of x_0 in Eq. (1.38), we obtain the complex solution of the differential Eq. (1.38) as

$$x = \frac{\frac{F_0}{m} e^{i(\omega t + \theta_0 + \beta)}}{i\left[\left(\omega_O^2 - \omega^2\right)^2 + 4\gamma^2\omega^2\right]^{\frac{1}{2}}} \tag{1.44}$$

This is the expression for a complex position. The real position will be the real part of the aforementioned equation. Hence, the real position of the forced harmonic oscillator is given by

$$x = \frac{F_0}{m\left[\left(\omega_0^2 - \omega^2\right)^2 + 4\gamma^2\omega^2\right]^{\frac{1}{2}}} \sin\left(\omega t + \theta_0 + \beta\right) \tag{1.45}$$

The general solution of the differential equation of motion (1.37) of a forced harmonic oscillation is

$$x = re^{-\gamma t} \cos\left(\omega_1 t + \theta\right) + \frac{F_0}{m\left[\left(\omega_0^2 - \omega^2\right)^2 + 4\gamma^2\omega^2\right]^{\frac{1}{2}}} \sin\left(\omega t + \theta_0 + \beta\right) \tag{1.46}$$

This above solution contains two constants namely r and θ which can be determined from the initial boundary conditions. The first term dies out exponentially in time and is called a transient term. After some time, the driven body starts oscillating with the frequency of the driving force or the applied force. This state is said to be the steady state of the forced vibrations. The second term is called the steady state term and oscillates with constant amplitude $\frac{F_0}{m\left[\left(\omega_0^2 - \omega^2\right)^2 + 4\gamma^2\omega^2\right]^{\frac{1}{2}}}$. In steady state, Eq. (1.46) becomes

$$x = \frac{F_0}{m\left[\left(\omega_0^2 - \omega^2\right)^2 + 4\gamma^2\omega^2\right]^{\frac{1}{2}}} \sin\left(\omega t + \theta_0 + \beta\right) \tag{1.47}$$

The phase of oscillation of the forced oscillator is $\left(\omega \mathrm{t} + \theta_0 + \beta\right)$ and that of the applied force is $\left(\omega t + \theta_0\right)$. Hence, the phase difference between oscillation and the applied force is β and is defined mathematically in Eq. (1.42) as

$$\tan^{-1} \frac{2\gamma\omega}{\omega_0^2 - \omega^2} = \frac{\pi}{2} - \beta$$

or $$\beta = \tan^{-1} \frac{\omega_0^2 - \omega^2}{2\gamma\omega} \tag{1.48}$$

This equation shows that the phase difference between oscillation and the applied force depends upon the frequency of the applied force and the damping coefficient of the medium.

1.4.1 Velocity of the forced harmonic oscillator

The complex velocity of the forced harmonic oscillator is obtained by differentiating Eq. (1.44) with respect to time as

$$v = i\omega \frac{\frac{F_0}{m} e^{i(\omega t+\theta_0+\beta)}}{i\left[\left(\omega_0^2-\omega^2\right)^2+4\gamma^2\omega^2\right]^{\frac{1}{2}}} = \omega \frac{\frac{F_0}{m} e^{i(\omega t+\theta_0+\beta)}}{\left[\left(\omega_0^2-\omega^2\right)^2+4\gamma^2\omega^2\right]^{\frac{1}{2}}}$$

The real velocity will be the real part of the aforementioned equation. Hence, the real velocity of the forced harmonic oscillator will be

$$v = \frac{\omega F_0}{m\left[\left(\omega_0^2-\omega^2\right)^2+4\gamma^2\omega^2\right]^{\frac{1}{2}}}\cos\left(\omega t+\theta_0+\beta\right) \tag{1.49}$$

1.4.2 Total energy of the forced harmonic oscillator

The instantaneous kinetic energy E_K of the forced harmonic oscillator is given as

$$E_K = \frac{1}{2}mv^2 .$$

Putting the value of v from Eq. (1.49) into the aforementioned equation, we get

$$E_K = \frac{1}{2}\frac{\omega^2 F_0^2}{m\left[\left(\omega_0^2-\omega^2\right)^2+4\gamma^2\omega^2\right]}\cos^2\left(\omega t+\theta_0+\beta\right) \tag{1.50}$$

The instantaneous potential energy E_P of the forced harmonic oscillator is given as

$$E_P = \frac{1}{2}kx^2$$

Putting the value of x from Eq. (1.47) into the aforementioned equation, we get

$$E_P = \frac{1}{2}m\omega_0^2 \times \frac{F_0^2}{m^2\left[\left(\omega_0^2-\omega^2\right)^2+4\gamma^2\omega^2\right]}\sin^2\left(\omega t+\theta_0+\beta\right)$$

or $$E_P = \frac{1}{2} \frac{\omega_0^2 F_0^2}{m\left[\left(\omega_0^2 - \omega^2\right)^2 + 4\gamma^2\omega^2\right]} \sin^2\left(\omega t + \theta_0 + \beta\right) \tag{1.51}$$

The total energy E of the forced harmonic oscillator will be given by

$$E = E_K + E_P$$

Putting the values of E_K and E_P from Eqs (1.50) and (1.51) into the aforementioned equation, we have

$$E = \frac{1}{2} \frac{\omega^2 F_0^2}{m\left[\left(\omega_0^2 - \omega^2\right)^2 + 4\gamma^2\omega^2\right]} \cos^2\left(\omega t + \theta_0 + \beta\right)$$

$$+ \frac{1}{2} \frac{\omega_0^2 F_0^2}{m\left[\left(\omega_0^2 - \omega^2\right)^2 + 4\gamma^2\omega^2\right]} \sin^2\left(\omega t + \theta_0 + \beta\right)$$

The total average energy $< E >$ of the forced harmonic oscillator over a complete cycle will be given by

$$< E > = \frac{1}{2} \frac{\omega^2 F_0^2}{m\left[\left(\omega_0^2 - \omega^2\right)^2 + 4\gamma^2\omega^2\right]} < \cos^2\left(\omega t + \theta_0 + \beta\right) >$$

$$+ \frac{1}{2} \frac{\omega_0^2 F_0^2}{m\left[\left(\omega_0^2 - \omega^2\right)^2 + 4\gamma^2\omega^2\right]} < \sin^2\left(\omega t + \theta_0 + \beta\right) >$$

or $$< E > = \frac{1}{2} \frac{\omega_0^2 F_0^2}{m\left[\left(\omega_0^2 - \omega^2\right)^2 + 4\gamma^2\omega^2\right]} \times \frac{1}{2}$$

$$+ \frac{1}{2} \frac{\omega_0^2 F_0^2}{m\left[\left(\omega_0^2 - \omega^2\right)^2 + 4\gamma^2\omega^2\right]} \times \frac{1}{2}$$

or $$<E>=\frac{1}{4}\frac{F_0^2}{m\left[\left(\omega_0^2-\omega^2\right)^2+4\gamma^2\omega^2\right]}\left(\omega^2+\omega_0^2\right) \tag{1.52}$$

1.4.3 Power of the forced harmonic oscillator

In steady state, the power dissipated by the oscillator due to a dissipative medium is compensated by the power absorption from the driving source. The power of a body 'P' moving under the action of force F with velocity v is defined as Fv. Therefore, the power dissipated by the forced harmonic oscillator will be given by

$$P = Fv$$

Putting the value of v from Eq. (1.49) and $F = F_0\cos(\omega t+\theta_0)$, which is the externally applied sinusoidally varying force, we get the power of the forced harmonic oscillator as

$$P=\frac{F_0^2}{m}\frac{\omega}{\left[\left(\omega_0^2-\omega^2\right)^2+4\gamma^2\omega^2\right]^{\frac{1}{2}}}\cos\left(\omega t+\theta_0\right)\cos\left(\omega t+\theta_0+\beta\right)$$

$$=\frac{F_0^2}{m}\frac{\omega}{\left[\left(\omega_0^2-\omega^2\right)^2+4\gamma^2\omega^2\right]^{\frac{1}{2}}}\cos\left(\omega t+\theta_0\right)$$

$$\times\left[\cos\left(\omega t+\theta_0\right)\cos\beta-\sin\left(\omega t+\theta_0\right)\sin\beta\right]$$

$$=\frac{F_0^2}{m}\frac{\omega}{\left[\left(\omega_0^2-\omega^2\right)^2+4\gamma^2\omega^2\right]^{\frac{1}{2}}}\left[\cos\beta\cos^2\left(\omega t+\theta_0\right)-\frac{1}{2}\sin 2\left(\omega t+\theta_0\right)\sin\beta\right]$$

or $$P=\frac{F_0^2}{m}\frac{\omega\cos\beta\cos^2\left(\omega t+\theta_0\right)}{\left[\left(\omega_0^2-\omega^2\right)^2+4\gamma^2\omega^2\right]^{\frac{1}{2}}}-\frac{F_0}{2m}\frac{\omega\sin\beta\sin 2\left(\omega t+\theta_0\right)}{\left[\left(\omega_0^2-\omega^2\right)^2+4\gamma^2\omega^2\right]^{\frac{1}{2}}} \tag{1.53}$$

Now the average value of $\cos^2(\omega t+\theta_0)$ and $\sin 2(\omega t+\theta_0)$ over a complete cycle of the applied force is $\frac{1}{2}$ and zero respectively. Therefore, the average value of power $<P>$ over a complete cycle will be

$$<P> = \frac{F_0^2}{2m}\frac{\omega\cos\beta}{\left[\left(\omega_0^2-\omega^2\right)^2+4\gamma^2\omega^2\right]^{\frac{1}{2}}} \tag{1.54}$$

Now either from Eq. (1.48) or from Fig. 1.6, we can have

$$\cos\beta = \frac{2\gamma\omega}{\left[\left(\omega_0^2-\omega^2\right)^2+4\gamma^2\omega^2\right]^{\frac{1}{2}}}$$

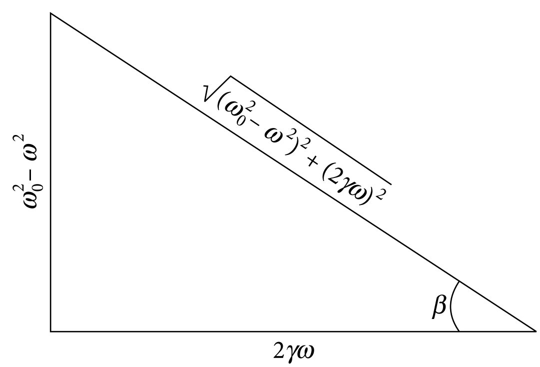

Figure 1.6 | Evaluation of trigonometric angles. This right triangle is drawn in accordance with Eq. (1.48)

Putting this value of $\cos\beta$ into Eq. (1.54), we get

$$<P> = \frac{F_0^2}{m}\frac{\gamma\omega^2}{\left[\left(\omega_0^2-\omega^2\right)^2+4\gamma^2\omega^2\right]^{\frac{1}{2}}} \tag{1.55}$$

This expression gives the amount of average power dissipated by the forced oscillator in the dissipative medium over a complete cycle of oscillation. In steady state, this must be equal to the power absorbed by the forced oscillator from the driving source.

Example 1.8

A ball of mass m performs undamped harmonic oscillations about the point $x = 0$ with natural frequency ω_0. At the moment $t = 0$, when the ball is in equilibrium position, a force $F_X = F_0 \cos \omega t$ along the x-axis was applied to it. Find the position of the ball at any time.

Solution

In this problem $\gamma = 0$ because the motion is undamped and $\theta_0 = 0$. So with the help of Eq. (1.46), we can determine the position of the ball by

$$x(t) = r\cos\omega t + \frac{\frac{F_0}{m}}{\omega_0^2 - \omega^2}\sin\left(\omega_0 t + \beta\right) \tag{A}$$

Putting $\beta = \frac{\pi}{2}$ into the Eq. (A), we get

$$x(t) = r\cos\omega t + \frac{\frac{F_0}{m}}{\omega_0^2 - \omega^2}\sin\left(\omega_0 t + \frac{\pi}{2}\right) = r\cos\omega t + \frac{\frac{F_0}{m}}{\omega_0^2 - \omega^2}\cos\omega_0 t \tag{B}$$

According to the problem, at $t = 0$, $x = 0$. Applying this boundary condition to (B), we get

$$x(0) = r\cos(\omega \times 0) + \frac{\frac{F_0}{m}}{\omega_0^2 - \omega^2}\cos\left(\omega_0 \times 0\right)$$

or $$r = -\frac{\frac{F_0}{m}}{\omega_0^2 - \omega^2}$$

Putting the value of r into (B), we get

$$x(t) = -\frac{\frac{F_0}{m}}{\omega_0^2 - \omega^2}\cos\omega t + \frac{\frac{F_0}{m}}{\omega_0^2 - \omega^2}\cos\omega_0 t$$

or $$x(t) = \frac{\frac{F_0}{m}}{\omega_0^2 - \omega^2}\left(\cos\omega_0 t - \cos\omega t\right)$$

1.5 Displacement Resonance

Forced vibrations occur if a system is continuously driven by an external agency. A simple example is a child's swing that is pushed on each downswing. Of special interest are systems undergoing harmonic oscillation and driven by sinusoidal force. This leads to the important phenomenon of resonance. Resonance occurs when the driving frequency approaches the natural frequency of free vibrations. The result is a rapid take-up of energy by the vibrating system, with a continuous growth of the vibration amplitude. A simple disturbance can set a harmonic oscillator into motion. Repeated disturbances can increase the amplitude of the oscillations if they are applied in phase with the natural frequency. Even a very small disturbance, repeated periodically at just the right frequency, can cause a very large amplitude motion to build up. In general, whenever a system capable of oscillation is acted upon by a periodic series of impulses having frequencies equal to or nearly equal to the natural frequencies of oscillation of the system, the system is set into oscillation with a relatively large amplitude. This phenomenon is known as resonance. Therefore we can define resonance as the phenomenon in which a body is set to violent vibrations by a strong periodic force whose frequency coincides exactly or nearly with the natural frequency of the body.

The amplitude of forced vibration in the steady state from Eq. (1.45) is

$$\frac{F_0}{m}\frac{1}{\sqrt{\left(\omega_0^2-\omega^2\right)^2+\left(2\gamma\omega\right)^2}}$$

and, as is obvious, it depends on the frequency of the driving force ω. This amplitude is maximum when the denominator is minimum, i.e., when

$$\frac{d}{d\omega}\left[\left(\omega_0^2-\omega^2\right)^2+\left(2\gamma\omega\right)^2\right]^{\frac{1}{2}}=0$$

or $$\omega_0^2-\omega^2=2\gamma^2$$

or $$\omega=\omega_0\sqrt{1-2\left(\frac{\gamma}{\omega_0}\right)^2} \tag{1.56}$$

Thus, the amplitude is maximum when ω is given by Eq. (1.56). The frequency at which resonance occurs (i.e., amplitude of oscillation become maximum) is called resonant frequency. Thus, Eq. (1.56) gives the expression for resonant frequency. When damping is extremely small (i.e., $\gamma\to 0$), the resonance occurs at a frequency very close to the natural frequency of the system (i.e., $\omega\to\omega_0$ when $\gamma\to 0$). The variation of amplitude with ω is depicted in Fig. 1.7.

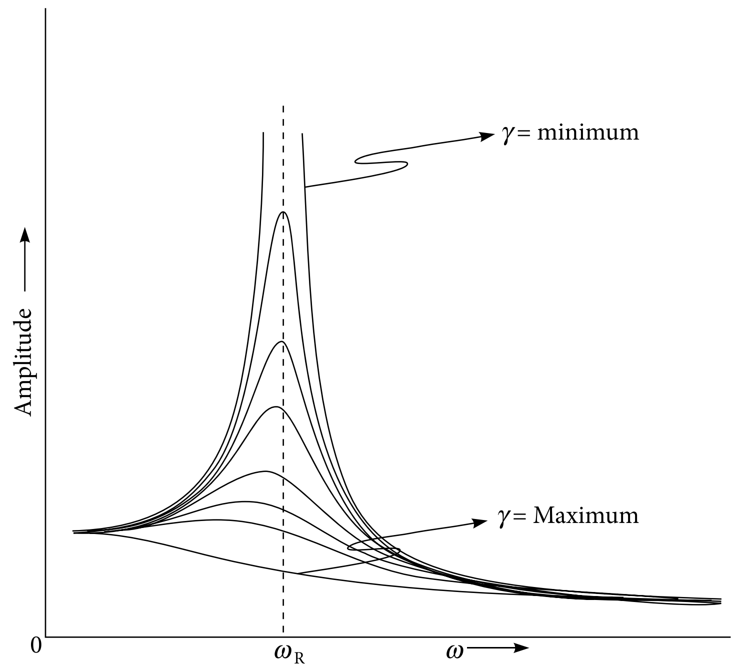

Figure 1.7 The variation of amplitude with frequency of the applied force for various values of γ. Observe that with increase of damping, the resonance occurs at smaller values of ω

It is interesting to observe that when damping decreases, maximum amplitude becomes very sharp and the amplitude falls off rapidly as we go away from the resonant frequency.

1.5.1 Resonant amplitude

The amplitude of the forced oscillation at resonance is called resonant amplitude R_R and the corresponding frequency is called resonant frequency ω_R. The expression for resonant frequency ω_R is obtained from Eq. (1.56) as

$$\omega_R = \omega_0 \sqrt{1 - 2\left(\frac{\gamma}{\omega_0}\right)^2} \tag{1.57}$$

The amplitude of forced vibration in the steady state from Eq. (1.45) is

$$\frac{F_0}{m} \frac{1}{\sqrt{\left(\omega_0^2 - \omega^2\right)^2 + \left(2\gamma\omega\right)^2}}$$

and hence, the expression for resonant amplitude R_R will be given by

$$R_R = \frac{F_0}{m}\frac{1}{\sqrt{\left(\omega_0^2 - \omega_R^2\right)^2 + 4\gamma^2\omega_R^2}}$$

Putting the value of ω_R from Eq. (1.57) into the aforementioned equation, we get

$$R_R = \frac{F_0}{m}\frac{1}{2\gamma\sqrt{\omega_0^2 - \gamma^2}} \tag{1.58}$$

If the damping coefficient of the medium is very small, we can neglect γ^2 in comparison to ω_0^2 and the amplitude of vibration at resonance is obtained as

$$R_R = \frac{F_0}{m}\frac{1}{2\gamma\omega_0} \tag{1.59}$$

Equations (1.58) or Eq. (1.59) shows that the amplitude at resonance becomes infinitely large when $\gamma \to 0$. Moreover, the same aforementioned equations shows that if the damping coefficient is very large (i.e., $\gamma \to \infty$), resonance cannot occur (i.e., $R_R \to 0$). The resonant frequency is inversely proportional to the damping coefficient for small damping.

1.5.2 Sharpness of resonance

The amplitude of vibration of an oscillatory system is maximum when the angular frequency of the sinusoidally varying applied force is equal to $\sqrt{\omega_0^2 - 2\gamma^2}$. The amplitude of vibration decreases when the angular frequency of the sinusoidally varying applied force is less than or more than $\sqrt{\omega_0^2 - 2\gamma^2}$. This has been depicted in Fig. 1.7. If the amplitude decreases rapidly for a slight departure of the frequency of the applied force from $\sqrt{\omega_0^2 - 2\gamma^2}$ the resonance is more sharp. If the amplitude decreases slowly for a large departure of the frequency of the applied force from $\sqrt{\omega_0^2 - 2\gamma^2}$, the resonance is less sharp. How rapidly the amplitude decreases on either side of $\sqrt{\omega_0^2 - 2\gamma^2}$ is represented by the sharpness of resonance.

The sharpness of resonance depends upon the damping coefficient. For small damping $(\gamma \to 0)$ at resonance, we can have

$$\omega_R \approx \omega_O$$

or $\quad \omega_R = \omega_0 \pm \gamma \qquad (\because \ \gamma$ is very small)

or $\quad \omega_1 = \omega_0 + \gamma$

and

$$\omega_2 = \omega_0 - \gamma$$

From the last two equations here, we have

$$\omega_1 - \omega_2 = 2\gamma = \frac{\omega_R}{Q} \tag{1.60}$$

Hence, $\omega_1 - \omega_2 = 2\gamma$ is the width of the resonance curve. The width of the resonance curve $\omega_1 - \omega_2$ is approximately proportional to the sharpness of resonance. Therefore, the sharpness of resonance is approximately proportional to the damping coefficient. More less the damping, more sharp is the resonance curve and vice versa. The more is the quality factor Q, the more sharp is the resonance curve.

1.5.3 Quality factor of a forced harmonic oscillator

According to relation (1.34), the quality factor Q of a forced harmonic oscillator can be defined as

$$Q = 2\pi \frac{\text{average energy stored per cycle}}{\text{average energy dissipated per cycle}} \tag{1.61}$$

or $$Q = 2\pi \frac{<E>}{T<P>},$$

T = time period of the applied force

or $$Q = \omega \frac{<E>}{<P>} \tag{1.62}$$

Putting the values of $<E>$ and $<P>$ from Eqs (1.52) and (1.55) respectively into Eq. (1.62) at resonance, we get

$$Q = \omega_R \frac{\dfrac{F_0^2\left(\omega_R^2 + \omega_0^2\right)}{4m\left[\left(\omega_0^2 - \omega_R^2\right)^2 + 4\gamma^2\omega_R^2\right]}}{\dfrac{\gamma\omega_R^2 F_0^2}{m\left[\left(\omega_R^2 - \omega_0^2\right)^2 + 4\gamma^2\omega^2\right]}}$$

or $$Q=\frac{\omega_R^2+\omega_0^2}{4\gamma\omega_R} \tag{1.63}$$

For small damping, $\omega_R \approx \omega_0$. Therefore, for small damping, the quality factor from Eq. (1.63) becomes

$$Q=\frac{2\omega_R^2}{4\gamma\omega_R}=\frac{\omega_R}{2\gamma} \tag{1.64}$$

Equations (1.63) and (1.64) show that larger the value of quality factor (i.e., lesser is γ), sharper is the resonance.

At displacement resonance from Eq. (1.58), we have

$$R_R=\frac{F_0}{m}\frac{1}{2\gamma\sqrt{\omega_0^2-\gamma^2}}$$

At low frequency from Eq. (1.47), we have

$$R_0=\frac{F_0}{m\omega_0^2}$$

or $$\frac{R_R}{R_0}=\frac{F_0}{m}\frac{1}{2\gamma\sqrt{\omega_0^2-\gamma^2}}\times\frac{m\omega_0^2}{F_0}=\frac{\omega_0}{2\gamma\left(1-\frac{\gamma^2}{\omega_0^2}\right)^{1/2}}$$

or $$\frac{R_R}{R_0}=\frac{Q}{\left(1-\frac{1}{4Q^2}\right)^{1/2}}\approx Q\left(1+\frac{1}{8Q^2}\right)\approx Q$$

The aforementioned equation predicts that the value of Q can be used as an amplification factor in different branches of science and technology.

1.5.4 Examples of resonance

i. When a tuning fork is struck and its stem is placed on the sonometer, the air inside the sonometer is set into resonant vibration.

ii. Soldiers marching over a suspension bridge are always advised to break their steps. If incidentally the frequency of the stepping of the soldiers coincide with the natural frequency of the bridge, violent oscillations set in, making the bridge to collapse. This was what happened to Tacoma Narrows Bridge (Washington, U.S.A.). In 1940, the newly constructed bridge collapsed by a mild storm just four months after it was built.

iii. Dancing of a small object placed on sound box.

iv. We are able to listen to a particular radio station because the tuned circuit in the radio set resonate at the frequency of the incoming electromagnetic wave which nearly coincide with its own natural frequency.

v. A stationary tuning fork vibrates when another vibrating tuning fork having a natural frequency equal to that of the stationary one is brought near it.

Example 1.9

Forced harmonic oscillations have the same displacement amplitudes at the natural frequencies ω_1 = 400/s and ω_2 = 800/s. Calculate the resonant frequency at which the displacement is maximum.

Solution

The amplitude of oscillation in case of forced vibration is given by

$$\frac{\frac{F_0}{m}}{\sqrt{\left(\omega^2-\omega_0^2\right)^2+\left(2\gamma\omega_0\right)^2}}$$

According to the problem,

$$\frac{\frac{F_0}{m}}{\sqrt{\left(\omega^2-\omega_1^2\right)^2+\left(2\gamma\omega_1\right)^2}}=\frac{\frac{F_0}{m}}{\sqrt{\left(\omega^2-\omega_2^2\right)^2+\left(2\gamma\omega_2\right)^2}}$$

or $2\omega^2-\omega_1^2-\omega_2^2=4\gamma^2$ (A)

From Eq. (1.56), the resonant frequency is given by

$$\omega_{\text{res}}=\sqrt{\omega^2-2\gamma^2}$$

Putting the value of $2\gamma^2$ from (A) into this equation, we have

$$\omega_{\text{res}}=\sqrt{\omega^2-\frac{2\omega^2-\omega_1^2-\omega_2^2}{2}}$$

or $$\omega_{\text{res}}=\sqrt{\frac{\omega_1^2+\omega_2^2}{2}}$$

According to the problem, ω_1 = 400/s and ω_2 = 800/s. Putting these values into the aforementioned equation, we get the resonant frequency at which the displacement is maximum as

$$\omega_{res} = \sqrt{\frac{400^2 + 800^2}{2}} = 632.46/\text{s} \quad \text{or} \quad \nu_{res} = \frac{\omega_{res}}{2\pi} = 100 \text{ Hz.}$$

1.6 Coupled Oscillators

In the section on simple harmonic oscillators, the motion of a single particle held in place by springs was considered. In this section, the motion of a group of particles bound by spring-like forces to one another is discussed.

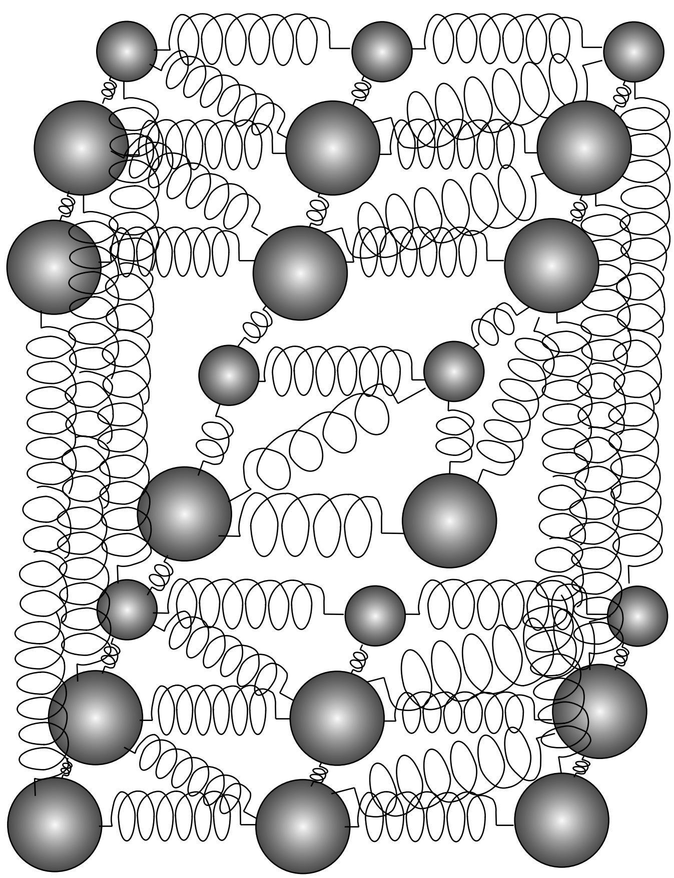

Figure 1.8 Cubic arrangement of atoms in a crystal as inferred from X-ray data. The spring like forces act between neighbouring atoms

Simple harmonic oscillators interacting with each other by any type of influence are called coupled oscillators. The understanding and manipulation of these phenomena has far-reaching implications in many fields of physics and engineering. For example, a system of particles held together by springs turns out to be a useful model of the behavior of atoms mutually bound in a crystalline solid. The atoms of a crystal are held in place by mutual forces of interaction that oppose any disturbance from equilibrium positions, similar to the forces in a spring (Fig. 1.8). For small displacements of the atoms, they behave mathematically just like spring forces – that is, they obey Hooke's law. Each atom is free to move in three dimensions rather than one and therefore, each atom added to a crystal adds three normal modes. In a typical crystal at ordinary temperature, all these modes are always excited by random thermal energy.

The lower-frequency, longer-wavelength modes may also be excited mechanically. These are called sound waves. The behavior of oscillatory electrical circuits inductively coupled to each other can be explained using the concepts of coupled oscillators.

To begin with a simple case, consider two particles in a line, as shown in Fig. 1.9. The system is a two-body coupled oscillator. Here the masses of the two bodies are same and equal to 'm' and the spring constants of the two springs are same and equal to 'k'. k_c is the spring constant of the coupling spring connecting the two bodies. For mathematical simplicity, the motion is restricted to one dimension only, i.e., along the x-axis. Even this elementary system is capable of surprising behavior. If one particle is held in place while the other is displaced and then both are released, the displaced particle immediately begins to execute simple harmonic oscillation.

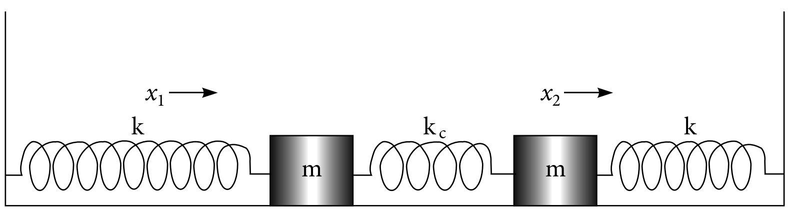

Figure 1.9 | A simple model of a two-body coupled oscillator.

This motion, by stretching the spring between the particles, starts to excite the second particle into motion. Gradually, the energy of motion passes from the first particle to the second particle and then the energy passes in the opposite direction. The motion of one mass depends upon the motion of the other mass. The system will oscillate with two degree of freedom and will have two natural frequencies. The mode of vibration in which the system vibrates with one frequency is called the normal mode of vibration. For each natural frequency, there corresponds a natural mode of vibration with a displacement configuration called the normal mode. Normal modes of vibrations are the free vibrations that depend only on the mass and stiffness of the system and how they are distributed.

To analyze the possible motions of the system, one may write the equations of motion for the two bodies in differential form. Again for mathematical simplicity, the system is made ideal, neglecting all dissipative forces like friction, air resistance, etc. The masses of the springs are also neglected. Assume that all the springs are initially at their natural lengths. We shall consider oscillations along the horizontal line (x-axis). This corresponds to longitudinal oscillations.

The force on each particle depends on two factors – first is on its displacement from its equilibrium position and second is on its distance from the other particle, because the spring between them stretches or compresses according to that distance. For this reason, the motions are coupled and the solution of each equation (i.e., the motion of each particle) depends on the solution of the other (i.e., the motion of the other).

Move the left mass a distance x_1 to the right and the right mass a distance x_2 to the right side. The coupling spring is therefore compressed by x_1 and stretched by x_2. The forces acting on the left mass are:

i. The force kx_1 towards the left side due to the stretching of the left spring by x_1.
ii. The force $k_C x_1$ to the left due to the compression of the coupling spring by x_1.
iii. The force $k_C x_2$ to the right due to the stretching of the coupling spring by x_2.

The resultant force F_1 acting on the left mass is

$$F_1 = -kx_1 - k_C x_1 + k_C x_2$$

Similarly, the resultant force F_2 acting on the right mass is

$$F_2 = -kx_2 - k_C x_2 + k_C x_1$$

Applying Newton's second law of motion to these equations, the differential equation of motion for both masses are respectively

$$m\frac{d^2 x_1}{dt^2} + kx_1 - k_C\left(x_2 - x_1\right) = 0 \tag{1.65}$$

$$m\frac{d^2 x_2}{dt^2} + kx_2 - k_C\left(x_1 - x_2\right) = 0 \tag{1.66}$$

Equations (1.64) and (1.66) can be respectively rewritten as

$$\frac{d^2 x_1}{dt^2} + \frac{k + k_C}{m}x_1 - \frac{k_C}{m}x_2 = 0 \tag{1.67}$$

$$\frac{d^2 x_2}{dt^2} + \frac{k + k_C}{m}x_2 - \frac{k_C}{m}x_1 = 0 \tag{1.68}$$

Addition and subtraction of Eqs (1.67) and (1.68) gives

$$\frac{d^2\left(x_1 + x_2\right)}{dt^2} + \frac{k}{m}\left(x_1 + x_2\right) = 0 \tag{1.69}$$

$$\frac{d^2(x_1 - x_2)}{dt^2} + \frac{k + 2k_C}{m}(x_1 - x_2) = 0 \tag{1.70}$$

Equations (1.69) and (1.70) indicate that $x_1 + x_2$ and $x_1 - x_2$ are the normal coordinates of the motion. Also $\sqrt{\frac{k}{m}}$ and $\sqrt{\frac{k+2k_c}{m}}$ are the normal frequencies of the system. It turns out that these normal frequencies are the frequencies at which the masses oscillate in their normal modes of vibration. This will be further explained later. With the following substitution into the aforementioned equations

$$q_1 = x_1 + x_2 \tag{1.71}$$

$$q_2 = x_1 - x_2 \tag{1.72}$$

$$\omega_1 = \sqrt{\frac{k}{m}} \tag{1.73}$$

$$\omega_2 = \sqrt{\frac{k + 2k_c}{m}} \tag{1.74}$$

we have

$$\frac{d^2 q_1}{dt^2} + \omega_1^2 q_1 = 0 \tag{1.75}$$

$$\frac{d^2 q_2}{dt^2} + \omega_2^2 q_2 = 0 \tag{1.76}$$

Now we have our equations in a much simpler form. In fact, these equations are exactly the same as that of the simple harmonic motion. The solutions are

$$q_1(t) = C_1' \cos \omega_1 t + C_2' \sin \omega_1 t \tag{1.77}$$

$$q_2(t) = C_3' \cos \omega_2 t + C_4' \sin \omega_2 t \tag{1.78}$$

From Eqs (1.71) and (1.72), we have

$$x_1 = \frac{q_1 + q_2}{2} \quad \text{and} \quad x_2 = \frac{q_1 - q_2}{2} \tag{1.79}$$

Thus, general solutions in terms of x_2 and x_1 are given as

$$x_1(t) = C_1 \cos\omega_1 t + C_2 \sin\omega_1 t + C_3 \cos\omega_2 t + C_4 \sin\omega_2 t \tag{1.80}$$

$$x_2(t) = C_1 \cos\omega_1 t + C_2 \sin\omega_1 t - C_3 \cos\omega_2 t - C_4 \sin\omega_2 t \tag{1.81}$$

Equations (1.80) and (1.81) show that motion of both masses is a superposition of two harmonic vibrations having angular frequencies ω_1 and ω_2. The constants C_1, C_2, C_3 and C_4 are evaluated from initial boundary conditions.

The coordinates q_1 and q_2 in terms of which the equations of motion take the form of a set of linear ordinary differential equations with constant coefficients involving just one of the coordinates, are called normal coordinates. A vibration (solution) involving only one dependent variable (normal coordinate) is called a normal mode of vibration and has its own normal frequency. In other words, we can define normal mode vibration as a vibration in which both masses harmonically oscillate at the same frequency. In such a normal mode, all the components of the system oscillate with the same normal frequency. Note that the normal mode corresponding to q_1 has a frequency ω_1, which is equal to the frequency of the individual oscillators. The normal mode corresponding to q_2 has a frequency ω_2, which is greater than the frequency of the individual oscillators. Since $\omega_2 > \omega_1$, the oscillation with frequency ω_1 is referred to as the normal mode of vibration of the coupled oscillator with lowest mode and the oscillation with frequency ω_2 is referred to as the normal mode of vibration of the coupled oscillator with highest mode.

Case 1: Normal mode of oscillation at higher frequency (anti-symmetric mode)

According to Eqs (1.80) and (1.81), the normal mode of vibration of higher frequency ω_2 occurs when $C_1 = 0$ and $C_2 = 0$. Under this condition, Eqs (1.80) and (1.81) become

$$x_1(t) = C_3 \cos\omega_2 t + C_4 \sin\omega_2 t \tag{1.82}$$

$$x_2(t) = -C_3 \cos\omega_2 t - C_4 \sin\omega_2 t \tag{1.83}$$

In Eqs (1.82) and (1.83), x_1 and x_2 are opposite in sign and equal in magnitude. Therefore, we conclude that in the normal mode of vibration of higher frequency ω_2, the two masses are vibrating in opposite directions or are out of phase. The normal mode of vibration of higher frequency is also known as the anti-symmetric mode.

Case 2: Normal mode of oscillation at lower frequency (symmetric mode)

Similarly, the normal mode of vibration at lower frequency ω_1 is obtained when $C_3 = 0$ and $C_4 = 0$. Under this condition, Eqs (1.80) and (1.81) become

$$x_1(t) = C_1 \cos\omega_1 t + C_2 \sin\omega_1 t \tag{1.84}$$

$$x_2(t) = C_1 \cos \omega_1 t + C_2 \sin \omega_1 t \tag{1.85}$$

In Eqs (1.84) and (1.85), x_1 and x_2 have the same sign and are equal in magnitude. Therefore, we conclude that in the normal mode of vibration of lower frequency ω_1, the two masses are vibrating in the same direction or are in phase. The normal mode of vibration of lower frequency is also known as the symmetric mode.

Linear combination of normal modes

Suppose the left-side mass is displaced through a distance '*r*' from its rest position keeping the right-side mass at rest. Both masses are released at time $t = 0$. Mathematically, this case can be described as

$$x_1(0) = r \quad \text{and} \quad \left.\frac{dx_1(t)}{dt}\right|_{t=0} = 0$$

$$x_2(0) = 0 \quad \text{and} \quad \left.\frac{dx_2(t)}{dt}\right|_{t=0} = 0$$

Putting these conditions into Eqs (1.80) and (1.81), we obtain

$$C_1 = \frac{r}{2},\ C_3 = \frac{r}{2},\ C_2 = 0,\ \text{and}\ C_4 = 0$$

Putting $C_1 = \frac{r}{2},\ C_3 = \frac{r}{2},\ C_2 = 0,\ \text{and}\ C_4 = 0$ into Eqs (1.80) and (1.81), we obtain

$$x_1(t) = \frac{r}{2}\left(\cos \omega_1 t + \cos \omega_2 t\right) = r \cos\left(\frac{\omega_2 + \omega_1}{2}\right) t \cos\left(\frac{\omega_2 - \omega_1}{2}\right) t \tag{1.86}$$

$$x_2(t) = \frac{r}{2}\left(\cos \omega_1 t - \cos \omega_2 t\right) = r \sin\left(\frac{\omega_2 + \omega_1}{2}\right) t \sin\left(\frac{\omega_2 - \omega_1}{2}\right) t \tag{1.87}$$

Unlike the case of pure normal modes, neither mass performs simple harmonic motion. The amplitude of the left-side mass is $r \cos\left(\frac{\omega_2 + \omega_1}{2}\right) t$, whereas amplitude of the right-side mass is $r \sin\left(\frac{\omega_2 + \omega_1}{2}\right) t$. Since there is a phase difference of $\pi / 2$, the crest of x_1 corresponds to the troughs of x_2. Each mass oscillates between the maximum amplitude '*r*' and an amplitude of zero This has been illustrated in Fig. 1.10

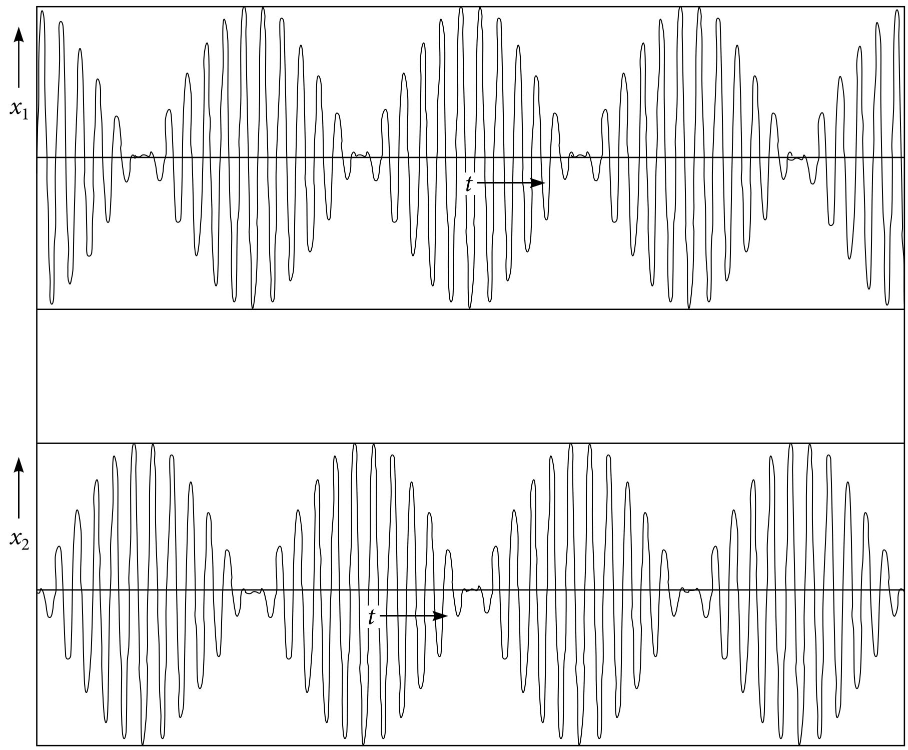

Figure 1.10 Linear combination of normal mode when $x_1(0) = r$, $x_2(0) = 0$ and the left-side mass is released from rest

If we give the system initial conditions of $x_1(0) = r_1$, $x_2(0) = r_2$ (i.e., two bodies have been displaced through distances of r_1 and r_2),

$$\frac{dx_1(0)}{dt} = 0 \text{ and } \frac{dx_2(0)}{dt} = 0$$

(i.e., the two masses are released from rest), we can see a similar behavior. This has been illustrated in Fig. 1.11.

However, neither mass's oscillations ever reach an amplitude of zero. This is because neither mass started at its equilibrium position. Here each mass oscillates between the maximum amplitude r_1 and the minimum amplitude r_2. The energy of the system is continuously oscillating between the two masses, while the total energy of the system remains constant. The mathematics of the problem is left as an exercise to the students.

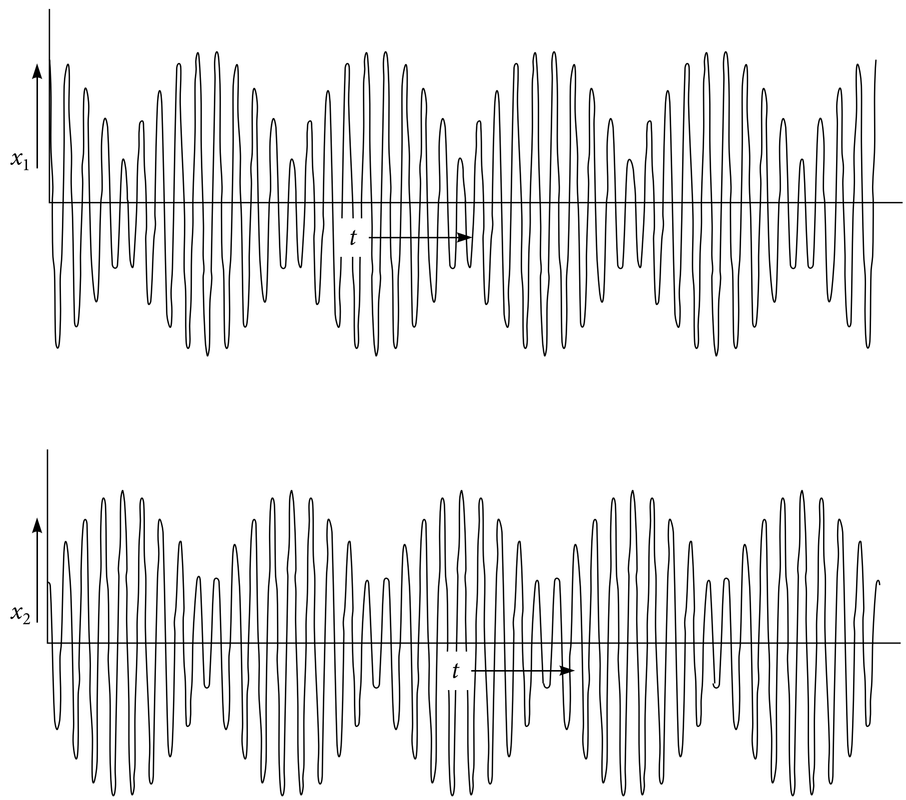

Figure 1.11 Linear combination of normal mode when $x_1(0) = r_1$, $x_2(0) = r_2$ and the two masses are released from rest

1.6.1 Experiment on a two-body coupled oscillator

An easy and interesting experiment can be performed on the coupled oscillator. A thick cotton string is fixed tightly between two rigid support separated by a distance of less than one meter. Two identical (i.e., same length and same mass) simple pendulums of length approximately half meter are taken – they can be named A and B. These two simple pendulums are knotted firmly to the tightly fixed string with a separation of around half meter. After knotting the two pendulums, the cotton string will sag slightly. Let the two pendulums become stationary. Without disturbing the other pendulum (B), oscillate one pendulum (A) in a plane perpendicular to the tightly fixed string.

Now observe the two pendulums. After some time, the stationary pendulum B will start oscillating with a slowly increasing amplitude and the oscillating pendulum A oscillates with decreasing amplitude. Again, after some time, pendulum A comes to rest/equilibrium position and pendulum B oscillates with the largest amplitude. Then the phenomenon will start to reverse. This process will continue for infinite time. Energy transfer takes place automatically from one pendulum to the other periodically with a certain fixed time period. However, practically, the two pendulums will come to rest due to the damping coefficient of the air medium and other practical difficulties. Interesting!

Students can do this experiment at home. The experiment can also be performed by taking more than two pendulums of different lengths and different masses. Naturally, the observation will be very much complicated, the degree of complexity depending upon the number of such pendulums chosen and their relative positions.

1.7 Analogy of Mechanical and Electrical Oscillations

It would be very useful to show the analogy of mechanical and electrical oscillations. The concepts developed earlier for mechanical oscillations can be obviously applied very well to electrical oscillations in circuits. The behavior of charge or currents in electric circuits can be described in terms of mechanical free, damped and forced harmonic oscillations. Suppose an electric circuit consists of a resistor with resistance R, an inductor with inductance L and a capacitor with capacitance C in series as shown in Fig. 1.12.

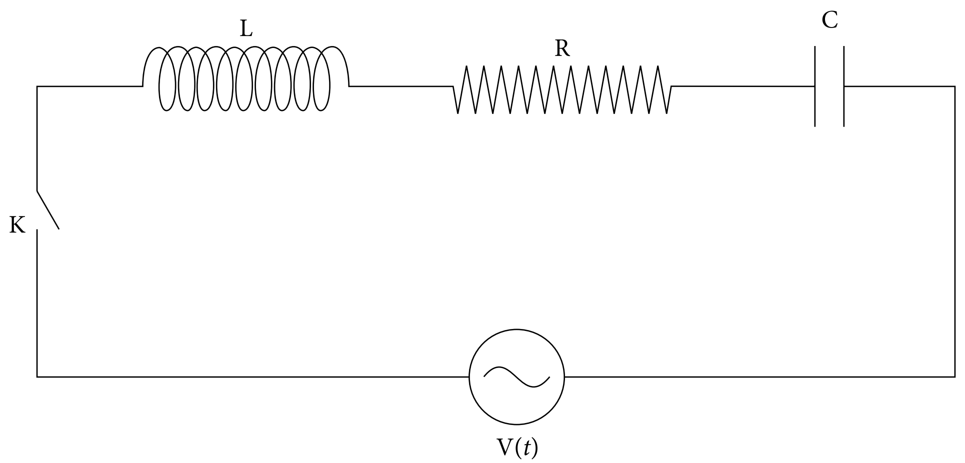

Figure 1.12 | LCR series circuit with applied voltage $V(t)$

If V_R, V_L and V_C are the instantaneous voltages across resistor, inductor, and capacitor respectively, then by applying Kirchhoff's voltage law to the circuit, we have

$$V_R + V_L + V_C = V(t) \tag{1.88}$$

Here

t = time

$V(t)$ = time varying applied voltage

$$V_R = iR = R\frac{dq}{dt} = \text{voltage across resistor}$$

$$V_L = L\frac{di}{dt} = L\frac{d^2q}{dt^2} = \text{voltage across inductor}$$

$$V_C = \frac{q}{C} = \frac{\int_{-\infty}^{t} i(\tau)d\tau}{C} = \text{voltage across capacitor}$$

q = charge

i = current

Thus, Eq. (1.88) can be re-written either in the form

$$L\frac{di}{dt} + Ri + \frac{\int_{-\infty}^{t} i(\tau)d\tau}{C} = V(t) \tag{1.89}$$

or in the form

$$L\frac{d^2q}{dt^2} + R\frac{dq}{dt} + \frac{q}{C} = V(t) \tag{1.90}$$

Differentiating Eq. (1.90) with respect to 't', we have

$$L\frac{d^2i}{dt^2} + R\frac{di}{dt} + \frac{i}{C} = \frac{dV(t)}{dt} \tag{1.91}$$

For a sinusoidally varying voltage source, like $V(t) = V_O \sin\omega t$, Eqs (1.90) and (1.91) becomes respectively

$$L\frac{d^2q}{dt^2} + R\frac{dq}{dt} + \frac{q}{C} = V_0 \sin\omega t \tag{1.92}$$

$$L\frac{d^2i}{dt^2} + R\frac{di}{dt} + \frac{i}{C} = V_0\omega\cos\omega t \tag{1.93}$$

Under different boundary conditions, these differential equations can be solved for the current and charges at any instant of time.

A glance at Eqs (1.90)–(1.93) along with that of mechanical problems shows the equivalence of electrical and mechanical oscillations as shown in the following table.

Equivalence of Electrical and Mechanical Oscillations

Mechanical Oscillation	Electrical Oscillation
Mass	Inductance
Damping coefficient	Resistance
Spring constant	Reciprocal of Capacitance
Applied Force	Applied voltage
Displacement	Charge
Velocity	Current
Potential Energy in spring = $\frac{1}{2}kx^2$	Energy in Capacitor = $\frac{1}{2}\frac{q^2}{C}$
Kinetic Energy = $\frac{1}{2}mv^2$	Energy in inductor = $\frac{1}{2}Li^2$

For equivalent expressions in electrical oscillations we have to replace the variables and other parameters according to the aforementioned table. In this way, we can get voltage equations from force equations as shown in the following text.

For example, let us discuss briefly the damped harmonic electrical oscillation produced when a charged capacitor is connected to a resistor and inductor in series as shown in Fig. 1.13.

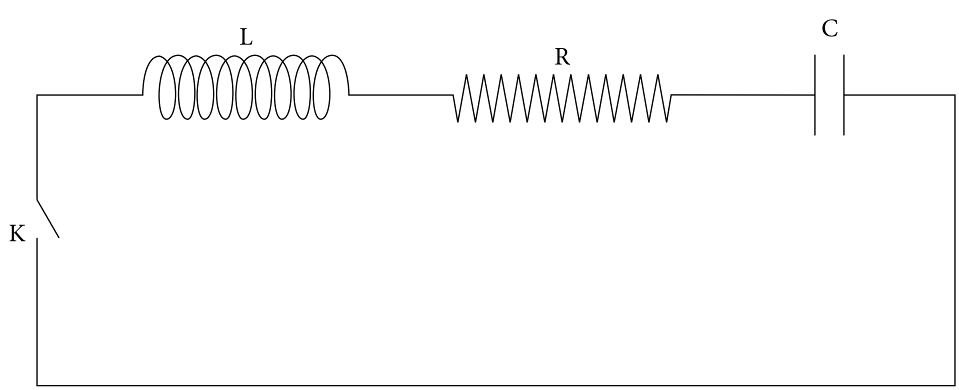

Figure 1.13 | LCR series circuit without applied voltage

In this case, since there is no externally applied voltage, from Eq. (1.90), we get

$$\frac{d^2q}{dt^2}+\frac{R}{L}\frac{dq}{dt}+\frac{1}{LC}q=0$$

Comparing this equation with Eq. (1.23) for $\frac{R^2}{4L^2}<\frac{1}{LC}$, we can have

$$q= q_0e^{-\gamma t}\cos\left(\omega_1 t+\theta\right) \tag{1.94}$$

Here $\omega_1=\sqrt{\omega_0^2-\gamma^2}$, $\omega_0=\frac{1}{\sqrt{LC}}$, $\gamma=\frac{R}{2L}$, $q_0e^{-\gamma t}$, and θ are the angular frequency, natural angular frequency, attenuation factor, instantaneous amplitude, and initial phase angle of oscillation respectively. The parameters q_0 and θ can be determined from initial boundary conditions.

Similarly, by comparing Eq. (1.93) with Eq. (1.36), we have

$$i=\frac{V_0\omega}{\sqrt{\left(\omega^2L-\frac{1}{C}\right)^2+R^2\omega^2}}\sin(\omega t+\beta)$$

with $\beta=\tan^{-1}\left(\frac{1-\omega^2LC}{RC\omega}\right)$

Example 1.10

An LCR circuit as shown in Fig. 1.13 contains an inductor of inductance 20.0 mH, a capacitor of capacitance 5.0 μF and a resistor of resistance 0.2 ohm. Calculate the angular frequency of oscillation. After how long a time will the charge oscillation decay to half of its initial amplitude. Assume the initial phase angle to be zero.

Solution

Data given are

$L = 20.0 \times 10^{-3}$ H

$C = 5.0 \times 10^{-6}$ F

$R = 0.2$ ohm

$$\frac{q}{q_0}=\frac{1}{2}$$

The angular frequency of the electrical oscillation is

$$\omega_1 = \sqrt{\frac{1}{LC} - \frac{R^2}{4L^2}} = \sqrt{1.0\times10^7 - 2.5}\ \text{rad/s} = 3162.3\ \text{rad/s}$$

$$\gamma = \frac{R}{2L} = 1.58\ \text{rad/s}$$

Putting the given conditions in Eq. (1.94), we get

$$\frac{1}{2} = e^{-1.58t}$$

or $t = \dfrac{\ln 2}{1.58}\ \text{s} = 0.44\ \text{s}$

Thus, the charge oscillation decays to half of its initial amplitude in 0.44 s.

1.8 Wave as a Periodic Variation Quantity in Space and Time

When a pebble is thrown into the still water of a pond, disturbance is created at the point where the pebble enters the water. The disturbance created thus is not confined to that point alone; it spreads out and reaches the boundary of the pond. The water does not move or flow towards the edge of the pond; only the disturbance moves. The water particles move up and down over a short distance in a vertical direction about their mean position as a result of which disturbance travels in a horizontal direction. At any time, the amount of disturbance is not the same at all points on the water surface. Also, at any point, the amount of disturbance varies with time. In non-technical language, we call this disturbance, water waves. This behavior is characteristic of all wave motions.

Now we can define the wave in the following way. A wave is defined as a disturbance that moves through a medium in such a manner that at any position, the displacement of the particles of the medium is a function of time while at any instant the displacement of the particles of the medium is a function of the position of the point. The medium as a whole does not move in the direction of the motion of the wave. The wave is also called a progressive wave or a travelling wave (see stationary waves).

1.8.1 Wave equation

Sinusoidal waves can be represented by an equation of the form

$$\psi(x,t) = r\sin(\omega t \pm kx) \quad \text{or} \quad \psi(x,t) = r\sin\left[k(vt \pm x)\right] \tag{1.95}$$

The '+' sign is used if the wave is travelling towards the left side ($-x$ direction) and the '–' sign is used if the wave is travelling towards the right-side ($+x$ direction).

Here

r = amplitude of the wave

ω = angular frequency of the wave

$k = \dfrac{2\pi}{\lambda}$ = wave number of the wave = magnitude of propagation vector.

v = phase speed of the wave $= \dfrac{\omega}{k}$

x and t = position and time functions.

The speed of the wave (wave speed) and the speeds of the particles of the medium (particle speed) are not the same. The particle speed is found out by differentiating the aforementioned equation. The particle speed is not the same for all the particles. The relation between maximum particle speed V_{max} and wave speed 'v' is given by

$$V_{max} = \frac{2\pi r}{\lambda} v \tag{1.96}$$

1.8.2 Wave equation in differential form

The general wave equation in one dimension is given by

$$\frac{\partial^2 \Psi(x,t)}{\partial x^2} = \frac{1}{v^2}\frac{\partial^2 \Psi(x,t)}{\partial t^2} + \frac{1}{\xi}\frac{\partial \Psi(x,t)}{\partial t} \tag{1.97}$$

and in three dimension, it is given by

$$\nabla^2 \Psi = \frac{1}{v^2}\frac{\partial^2 \Psi}{\partial t^2} + \frac{1}{\xi}\frac{\partial \Psi}{\partial t}$$

or $$\nabla^2 \Psi - \frac{1}{v^2}\frac{\partial^2 \Psi}{\partial t^2} - \frac{1}{\xi}\frac{\partial \Psi}{\partial t} = 0 \tag{1.98}$$

where $\nabla^2 = \dfrac{\partial^2}{\partial x^2} + \dfrac{\partial^2}{\partial y^2} + \dfrac{\partial^2}{\partial z^2}$ and is called a Laplacian operator. In these equations, ξ called diffusivity, takes care of the energy loss of the wave in the medium. Here, Ψ is called wave function and v is the speed of the wave. The wave function plays the role of disturbance in wave propagation. It contains all the information regarding the wave propagation.

The solution of this wave Eq. (1.97) gives the direction of propagation, the wave speed and many other properties of waves. In case of a stretched string, Ψ is the displacement of the string from the x-axis; in case of a sound wave, Ψ is the pressure difference; in case of light wave, Ψ is either an electric field or a magnetic field. Solutions of this differential wave equation are of many kinds, reflecting a variety of waves that can occur.

1.9 Longitudinal and Transverse Waves

Depending upon the direction of oscillation of the particles of the medium, progressive waves can be divided into two categories of waves; one is longitudinal waves and the other is transverse waves.

1.9.1 Longitudinal waves

A longitudinal wave is defined as a wave in which particles of the medium oscillate in the direction of propagation of the wave. A beautiful example of a longitudinal wave is the sound wave. Sound waves propagate in the air medium. In the propagation of sound waves, air particles oscillate (but do not move) about their mean position in the direction in which sound travels. In this case, air particles get compressed at one region and at the adjacent region, air particles get rarefied. This process goes on alternately. The two compression and rarefaction regions are pictorially represented in Fig. 1.14.

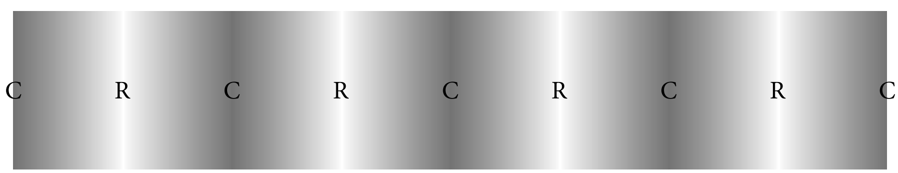

Figure 1.14 Propagation of longitudinal waves. *C* represents a compression region and *R* represents a rarefaction region

1.9.2 Transverse waves

The transverse wave is defined as the wave in which particles of the medium oscillate in a direction perpendicular (i.e., transverse) to the direction of propagation of the wave. One of the visible examples of a transverse wave is the water wave. Water wave propagates in the water medium. In the propagation of a water wave, water particles oscillate about their mean position in a direction perpendicular to the direction in which the water wave travels.

In this case, water particles get raised above the normal level of the water surface at one region and at the adjacent region, water particles get depressed below the normal level of the water surface. The raised region is called 'crest' and the depressed region is called 'trough'. This process goes on alternately and is represented in Fig. 1.15.

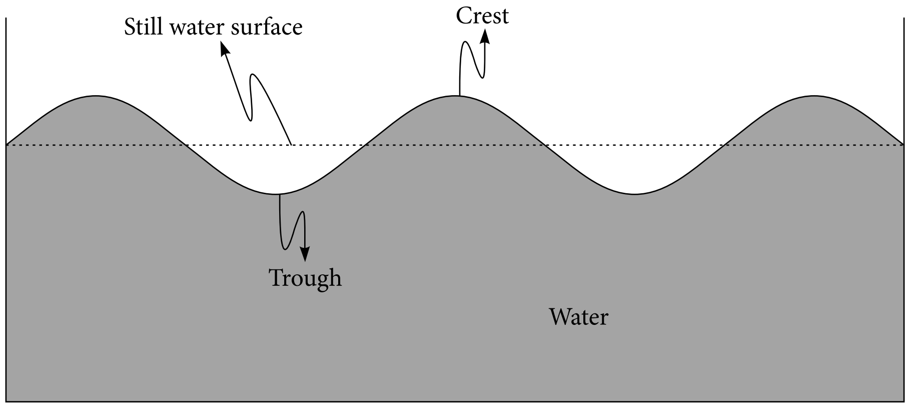

Figure 1.15 | Water waves or ripples

All electromagnetic waves belong to the transverse wave category. In this case, the electromagnetic field oscillates and no material medium is necessary.

Example 1.11

The equation of a wave moving along a string is $\Psi = 4 \sin \pi(0.04x - 8t)$. Here Ψ, x are in cm and t is in seconds. Find the amplitude, frequency, and speed of the wave. Also calculate the phase difference between two points situated 25 cm apart.

Solution

The given equation is $\Psi = 4\sin\pi(0.04x - 8t) = 4\sin 2\pi(0.02x - 4t)$

Comparing this equation with the standard equation $\Psi = r\sin 2\pi\left(\frac{x}{\lambda} - \frac{t}{T}\right)$, we get $r = 4$ cm, $\frac{1}{\lambda} = 0.02/\text{cm}$, and $\frac{1}{T} = 4/\text{s}$. Thus, we have amplitude $r = 4$ cms, wavelength $\lambda = 50$ cms, time period $T = 0.25$ s and frequency $\frac{1}{T} = 4$ Hz. The velocity of the wave is given by

Speed = frequency x. Wavelength = 4 × 50 cm/s = 200 cm/s

The phase difference between the two points is given by

$$\text{Phase difference} = \frac{2\pi}{\lambda} \times \text{path difference}$$

$$= \frac{2\pi}{50} \times 25 = \pi \text{ radian}$$

1.9.3 Difference between longitudinal waves and transverse waves

We can list the differences between longitudinal and transverse waves by having the mental picture of a sound wave and a water wave in our mind.

	Longitudinal wave	Transverse wave
1	Here, the particles oscillate about their mean position in the direction of propagation of the wave	Here, the particles oscillate about their mean position in the direction perpendicular to the direction of propagation of the wave
2	The wave travels in the form of compression and rarefaction. One compression and one rarefaction form a wave.	The wave travels in the form of crest and troughs. One crests and one trough form a wave.
3	The production of this type of wave is possible in all type of media	The production of this type of wave is possible in any type of media which have elasticity.
4	The longitudinal wave cannot be polarized	The transverse wave can be polarized
5	Examples: sound wave, seismic primary waves, shock waves in air (shock waves are created by violent explosion), etc.	Examples: All type of electromagnetic waves, waters waves, seismic shear waves, seismic Love waves, etc.

1.9.4 Characteristics of progressive waves

The following are the characteristics of progressive waves.

i. A progressive wave is the disturbance produced in the medium by the repeated periodic motion of the particles of the medium, motion being handed over from particle to particle.

ii. It is the progressive wave which advances. The particles of the medium do not travel but oscillate about their mean position.

iii. There is a continuous phase difference between the particles of the medium when progressive wave advances.

iv. The velocity of the progressive wave is uniform through out the medium.

v. When a progressive wave propagates, each particle of the medium oscillates about their mean position with the same amplitude and time period. The velocity of the particle is different at different points. It is maximum at the mean position and zero at the extreme position.

vi. The particle velocity is different from the wave velocity.

vii. There is transmission of energy across every plane in the direction of propagation waves

1.10 Stationary Waves

A stationary wave is defined as a wave produced by the superposition of two identical progressive waves (i.e., progressive waves having the same wavelength, the same time period, the same frequency and the same speed) propagating through a medium in a line but in opposite directions. The new wave produced by this method is called a stationary wave because there is no transmission of energy across any plane. That means stationary waves do not carry energy. There are certain points in the medium at which particles of the medium are permanently at rest. These points are called nodes. There are certain points in the medium at which particles of the medium are oscillating with maximum amplitude. These points are called anti nodes. The stationary wave can not carry energy because the particles of the medium at the nodes are permanently at rest without having any energy. Stationary waves are also called standing waves.

1.10.1 Formation of stationary waves

A stationary wave is formed when the incident wave and the corresponding wave reflected by a rigid wall are superimposed on each other. In Fig. 1.16, the displacement graphs of both incident and reflected waves are plotted.

The incident wave is supposed to be travelling from left to right. The reflected wave (Fig. 1.17) is travelling from right to left. These incident and reflected waves superimpose on each other and thus give rise to stationary waves. The continuous curve in 3 of Fig. 1.16 depicts the resultant stationary waves. The mode of stationary waves depends upon the phase difference of the two waves. The nature of stationary waves is shown graphically in Fig. 1.17 at different intervals. Here, the incident wave (I) is represented by dotted curves and the reflected wave (II) is represented by dashed line curves.

Case 1:

At $t = 0$, the second crest of the incident wave coincides with the trough of the reflected wave. Since the two waves have equal amplitude, the resultant stationary wave is a straight line represented by a solid line. At all the points P_1, P_2, P_3, P_4, P_5, P_6, P_7, P_8, P_9, the particles of the medium have zero amplitude. See Fig. 1.17(a)

Case 2:

At $t = T/4$, the second incident and reflected wave have individually advanced towards each other by a distance of $\lambda/4$ and the relative displacement between them is $\lambda/2$. $\left(\frac{\lambda}{4}+\frac{\lambda}{4}=\frac{\lambda}{2}\right)$. As a result of this, the crest of the incident wave coincides with the crest of the reflected wave giving a resultant stationary wave of maximum amplitude. Points P_1, P_3, P_5, P_7 and P_9 acquire maximum displacements while the points P_2, P_4, P_6, and P_8 have zero amplitude. See Fig. 1.17(b).

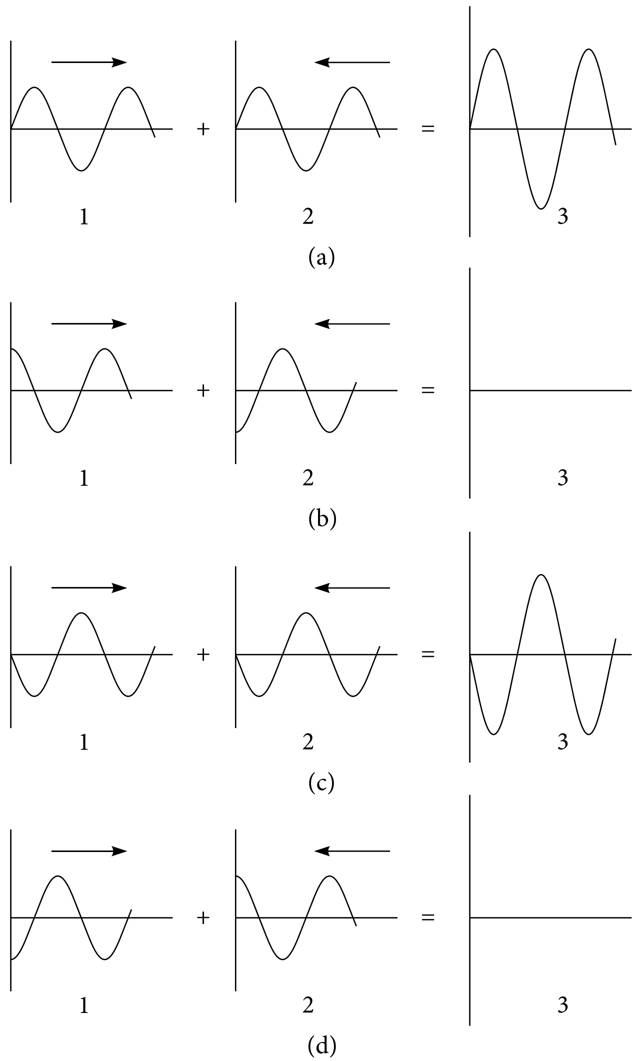

Figure 1.16 Stationary waves as superposition of right and left going waves. 1 and 2 are components, 3 the resultant

Case 3:

At $t = T/2$ second, the incident and the reflected wave have individually advanced towards each other by a distance of $\lambda/2$ and the relative displacement between them is $\lambda \left(\frac{\lambda}{2} + \frac{\lambda}{2} = \lambda\right)$. As a result, the crest of the incident wave coincides with the trough of the reflected wave giving a resultant stationary wave of zero amplitude. In this case, again, the resultant stationary wave is a straight line. See Fig. 1.17(c).

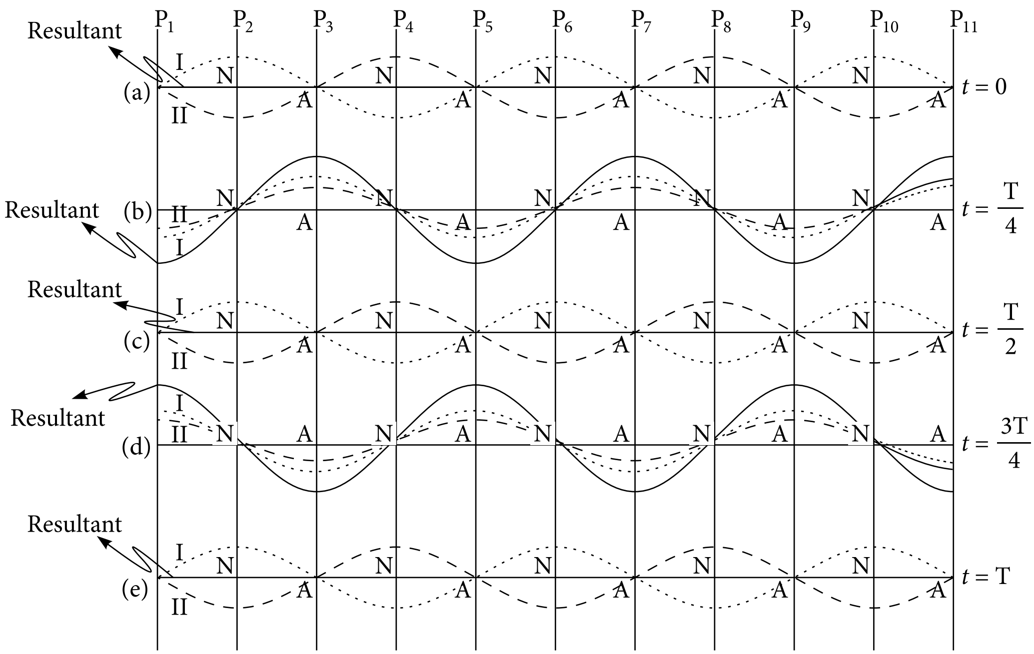

Figure 1.17 Graphical explanation for the formation of stationary waves. The dotted curve and the dashed line curve represent the incident and reflected curves respectively. The continuous curve is the stationary wave. Ns are the positions of nodes and As are the positions of antinodes

Case 4:

At $t = 3T/4$ second, the incident and reflected wave have individually advanced towards each other by a distance of $3\lambda/4$ and the relative displacement between them is

$$\lambda 3/2. \left(\frac{3\lambda}{4} + \frac{3\lambda}{4} = \frac{3\lambda}{2}\right).$$

As a result of this, the trough of the incident wave coincides with the trough of the reflected wave giving a resultant stationary wave of maximum amplitude but opposite to that of Case 2. Points P_1, P_3, P_5, P_7 and P_9 acquire maximum displacements while the points P_2, P_4, P_6, and P_8 have zero amplitude. See Fig. 1.17(d).

Case 5:

At $t = T$ second, the crest of the incident wave coincides with the trough of the reflected wave. Since the two waves have equal amplitude, the resultant stationary wave is a straight line represented by a solid line. At all the points, P_1, P_2, P_3, P_4, P_5, P_6, P_7, P_8, P_9, the particles of the medium have zero amplitude. See Fig. 1.17(e).

1.10.2 Characteristics of stationary waves

The followings are the characteristics of stationary waves.

i. A stationary wave is produced when two identical progressive waves (i.e., waves having the same wavelength, the same time period, the same frequency, and the same speed) travelling in the same line but in opposite directions are superimposed.

ii. The points at which particles are not oscillating are called nodes.

iii. The points at which particles are oscillating with maximum displacements are called anti nodes

iv. The distance between any two consecutive nodes is half the wavelength.

v. The distance between any two consecutive anti nodes is half the wavelength

vi. The distance between any two consecutive nodes and anti nodes is one-fourth of the wavelength.

vii. Except at nodes, each particle of the medium executes simple harmonic oscillation about its mean position with a time period equal to the time period of the wave motion.

viii. Amplitudes of all the particles of the medium are not the same. It is zero at the node and maximum at the anti node.

1.10.3 Differences between progressive and stationary waves

	Progressive waves	Stationary waves
1	A progressive wave is produced due to the oscillation of the particles of the medium.	A stationary wave is produced when two identical progressive waves travelling in the same line but opposite directions are superimposed.
2	The waves move with a velocity depending upon the properties of the medium.	The waves remain stationary and do not move.
3	Each particle of the medium executes periodic motion about their mean position with the same amplitude.	Except the node, all the particles of the medium execute SHO with varying amplitude.
4	There is a continuous change of phase from particle to particle.	All the particles between two consecutive nodes are at the same phase, but differ in phase by π from those in the preceding as well as succeeding similar segments.
5	At any instant all the particles do not come together in the mean position, they pass their mean position in succession but with the same velocity.	All the particles pass their mean position at a time, but with different velocities.

6	Each particle of the medium undergoes similar change of pressure and density	There is no change of pressure and densities at the antinodes while there is maximum change of pressure and densities at the nodes.
7	There is transmission of energy across every plane in the direction of propagation of waves.	There is no flow of energy across any plane.
8	A complete wavelength contains a compression and rarefaction in the case of longitudinal waves and crest and trough in the case of transverse waves.	The wavelength is the distance between two alternate nodes and anti nodes.
9	Compression and rarefaction move from point to point throughout the medium.	The compression and rarefaction do not move from point to point; they simply appear at and disappear at certain equidistance fixed points.
10	No particle of the medium is permanently at rest.	Particles at the nodes are permanently at rest.
11	The equation of a progressive wave is given by $\psi(x,t) = r\sin\ (kx \mp \omega t)$	The equation of a stationary wave is given by $\psi(x,t) = 2r\ \sin kx\ \cos\omega t$

1.11 Reflection of a Wave at the Boundary of Two Media

When a wave is allowed to fall on the interface separating two media, the wave is returned to the same medium completely or partially. This phenomenon is called reflection. Reflection is defined as the phenomenon by virtue of which wave (actually energy contained in the wave) is sent back to the same medium by an interface separating the two media.

1.11.1 Reflection of transverse waves

Case 1: Passing from rarer medium to denser medium

Let a transverse wave travel along a string fixed to a point 'P' on the rigid wall. Suppose a crest is formed in the string just before the wall. The string will exert an upward force on the point P as a result of which a downward force of same magnitude will act (Newton's third law) on the string at the point 'P'. The result will be the formation of a trough just before the point 'P'. See Fig. 1.18. Therefore, there is a phase change π when a transverse wave is reflected back by the surface of a denser medium.

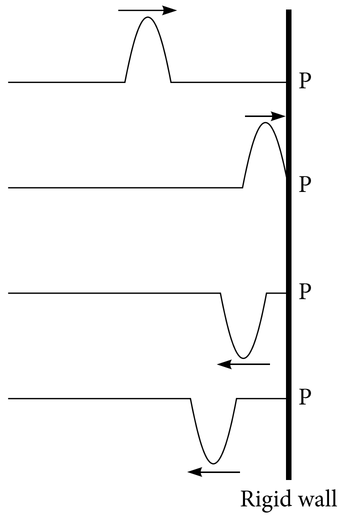

Figure 1.18 | Reflection of transverse waves from a rigid (hard) wall

Case 2: Passing from denser medium to rarer medium

Let a transverse wave travel along a string fixed to a point 'P' on a mass-less ring capable of sliding freely in the vertical direction without any friction on the rigid support. Suppose a crest is formed in the string just before the support. The string will exert an upward force on the point P as a result of which the ring will slide in the upward direction. The result will be the formation of a crest just before the point 'P'. See Fig. 1.19. Therefore, there is no phase change when a transverse wave is reflected back by the surface of a rarer medium.

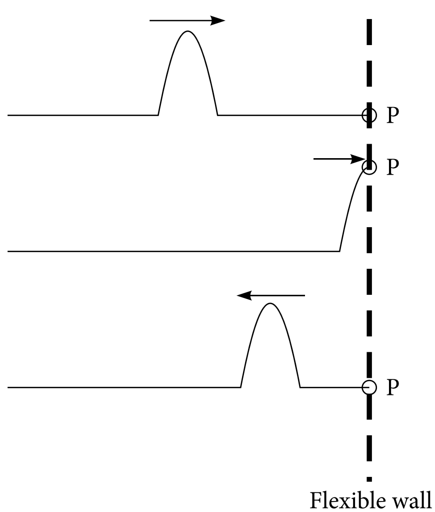

Figure 1.19 | Reflection of transverse waves from a flexible (soft) wall

1.11.2 Reflection of longitudinal waves

Case 1: Passing from rarer medium to denser medium

Longitudinal waves propagate by means of compression and rarefaction. Let a longitudinal wave travel towards the rigid wall, i.e., toward the interface separating the rarer medium from the denser medium. (imagine the propagation of sound waves). Suppose a compression region is formed just before the rigid wall. The compressed particles will rebound in a group forming a compression region in the vicinity of the wall. Thus, compression is reflected back as compression. The same is the fate of the rarefaction. Therefore, a wave of compression travelling from a rarer to a denser medium is reflected as a wave of compression while rarefaction is reflected as rarefaction. That means there is no phase change when a longitudinal wave is reflected back by the surface of a denser medium. See Fig. 1.20.

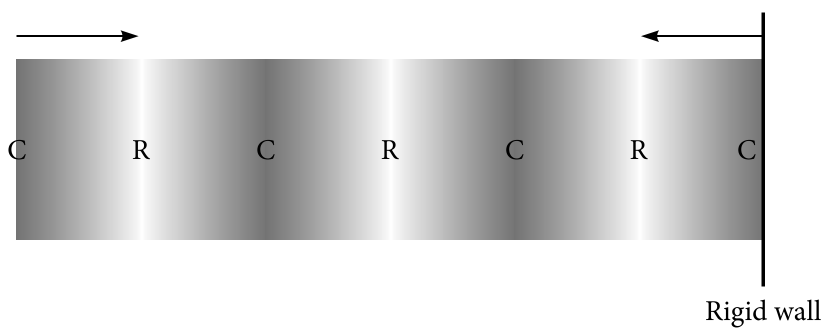

Figure 1.20 | Reflection of longitudinal waves from a rigid (hard) wall

Case 2: Passing from denser medium to rarer medium

Let a longitudinal wave travel towards the interface separating a denser medium from a rarer medium. Suppose a compression region is formed just before the interface. The compression region presses the particles of the rarer medium which yield to it and move forward with higher speed, leaving a rarefaction behind. This rarefaction travels back as a reflected wave. Thus, compression is reflected back as rarefaction. The same is the fate of the rarefaction. Therefore, a wave of compression travelling from a denser to a rarer medium is reflected as a wave of rarefaction while rarefaction is reflected as compression. That means that there is phase change of π when a longitudinal wave is reflected back by the surface of a rarer medium. See Fig. 1.21.

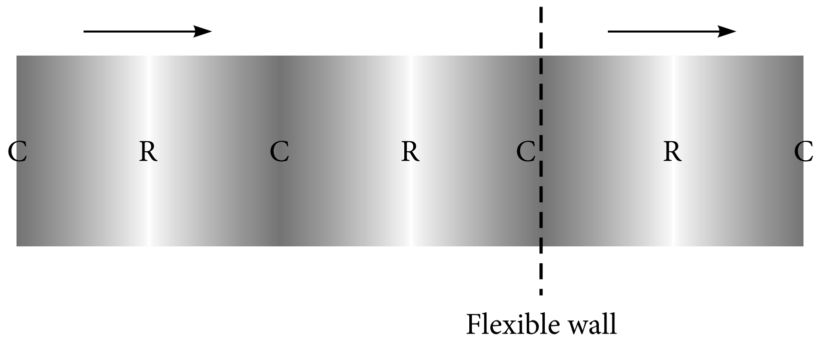

Figure 1.21 | Reflection of transverse waves from a flexible (soft) wall

1.12 Refraction of a Wave at the Boundary of Two Media

When a wave is allowed to fall on the interface separating two media, the wave is transmitted to the other medium completely or partially. This phenomenon is called refraction. Refraction is defined as the phenomenon by virtue of which wave (actually energy contained in the wave) is transmitted to the other medium through the interface separating the two media. The speed and wavelength of waves change in refraction.

1.12.1 Refraction of transverse waves

Case 1: Passing from rarer medium to denser medium

Let a transverse wave travel along a thin string attached to a thick string at a point 'P' on the interface. Suppose a crest is formed in the thin string just before the interface. The thin string will exert an upward force on the point P as a result of which a downward force of the same magnitude will act (Newton's third law) on the light string. The result will be the formation of a trough just before the interface and a crest just after the interface. See Fig. 1.22. Hence, a crest is refracted or transmitted as a crest. Therefore, there is no change of phase when a transverse wave is refracted going from a rarer medium to a denser medium.

Case 2: Passing from denser medium to rarer medium

Let a transverse wave travel along a thick string attached to a thin string at a point 'P' on the interface. Suppose a crest is formed in the thick string just before the interface. The thick string will exert an upward force on the point P as a result of which light string will yield to the upward force resulting in the formation of a crest just after and before the interface. See Fig. 1.23. Hence, a crest is refracted or transmitted as a crest. Therefore, there is no change of phase when a transverse wave is refracted going from a denser medium to a rarer medium.

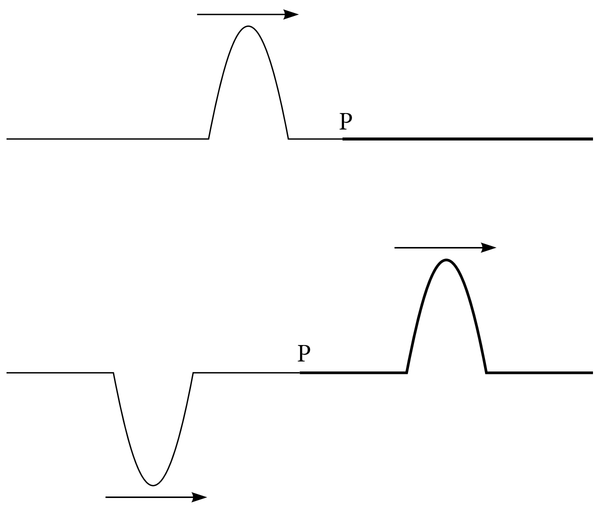

Figure 1.22 | Refraction of transverse waves through a flexible (soft) wall

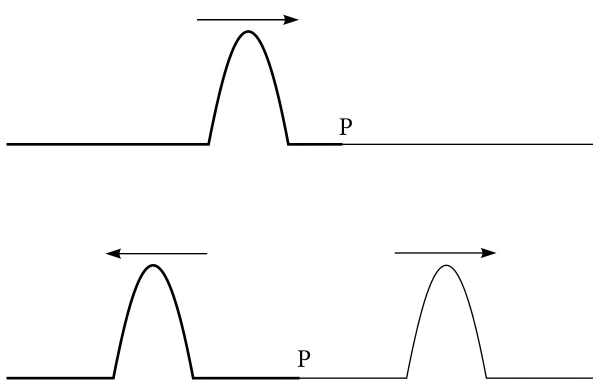

Figure 1.23 | Refraction of transverse waves through a rigid (hard) wall

1.12.2 Refraction of longitudinal waves

Case 1: Passing from rarer medium to denser medium

As mentioned earlier, longitudinal waves propagate in the form of compression and rarefaction. Let a longitudinal wave travel towards the rigid wall, i.e., toward the interface separating a rarer medium from a denser medium. Suppose a compression region is formed just before the rigid wall. See Fig. 1.24.

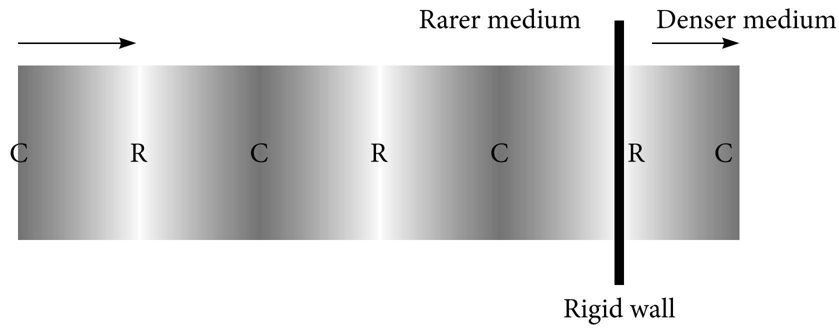

Figure 1.24 | Refraction of longitudinal waves through a rigid (hard) wall

The compressed particles will strike the wall in a group forming a compression region in the vicinity of the back of the wall. Thus, compression is refracted or transmitted as compression. The same is the fate of the rarefaction. Therefore, a wave of compression travelling from the rarer to the denser medium is refracted as a wave of compression while rarefaction is refracted as rarefaction. That means there is no phase change when a longitudinal wave is transmitted or refracted from a rarer medium to a denser medium.

Case 2: Passing from denser medium to rarer medium

Let a longitudinal wave travel towards the interface separating a denser medium from a rarer medium. Suppose a compression region is formed just before the interface. The compression region presses the particles of the rarer medium to replace them. Thus, compression is refracted as compression. The same is the fate of the rarefaction. Therefore, a wave of compression travelling from the denser to the rarer medium is refracted as a wave of compression while rarefaction is refracted as rarefaction. That means there is no change of phase when a longitudinal wave is refracted or transmitted from a denser medium and a rarer medium. See Fig. 1.25.

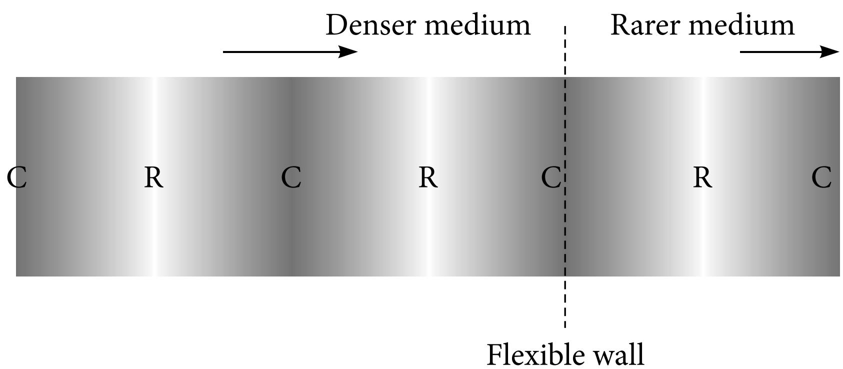

Figure 1.25 | Refraction of longitudinal waves through a flexible (soft) wall

Example 1.12

The equation of a stationary wave in a medium is given as

$$\Psi = 6\cos\frac{2\pi x}{13}\sin 20\pi t.$$

Here Ψ and x are in cm and t in seconds. Calculate the amplitude, wavelength, and velocity of the two component waves. Determine the distance between two consecutive nodes and anti-nodes. What is the velocity of a particle at a distance 13 cms at time 0.2 s?

Solution

The given equation

$$\Psi = 6\cos\frac{2\pi x}{13}\sin 20\pi t$$

can be written as

$$\Psi = 2\times 3\cos\frac{2\pi x}{13}\sin 2\pi\times 10t$$

Comparing this equation with the standard stationary wave equation

$$\Psi = 2r\cos\frac{2\pi x}{\lambda}\sin 2\pi\frac{t}{T},$$

we get

amplitude = 3 cms, wavelength λ = 13 cm time period T = 0.1 s, frequency = 10/s,
velocity = wavelength × frequency = 13 × 10 cm/s = 130 cm/s.

The distance between two consecutive nodes and anti nodes = 13 cms/2 = 6.5 cm.
The particle speed is given by

$$\frac{d\Psi}{dt} = \frac{d}{dt}6\cos\frac{2\pi x}{3}\sin 20\pi t = 6\cos\frac{2\pi x}{13}\cos 20\pi t\times 20\pi$$

So the particle speed at x = 13 cm and t = 0.2 s will be = $6\times\cos 2\pi\times\cos 4\pi\times 20\pi = 120\pi$ cm/s

Example 1.13

The vibration of a string 40 cms in length fixed at both ends are given by the equation

$$\Psi = 4\sin\frac{4\pi x}{25}\cos(4\pi t).$$

Here Ψ, x are in cm and t in seconds. What is the maximum displacement of a particle at 2.5 cm? Write down the equation of the component waves whose superposition gives this stationary wave.

Solution

The given equation

$$\Psi = 4\sin\frac{4\pi x}{25}\cos(4\pi t)$$

can be written as

$$\Psi = 2\times 2\sin\frac{2\times 2\pi x}{25}\cos(2\pi\times 2t)$$

Comparing this equation with a standard equation of the form

$$\Psi = 2r\sin\frac{2\pi x}{\lambda}\cos\left(2\pi\frac{t}{T}\right),$$

we have amplitude = 2 cm, wavelength = 25/2 = 12.5 cm, time period = 1/2 = 0.5 s, frequency = 2/s, speed = 12.5 × 2 = 25 cm/s

The maximum displacement of the particle of the medium is

$$\Psi_{max} = 4\sin\frac{4\pi x}{25}.$$

The maximum displacement of a particle at x = 2.5 cms is

$$4\times\sin\left(\frac{4\pi}{25}\times 2.5\right) = 4\times 0.95 = 3.80 \text{ cm}.$$

The equations of the component waves are

$$\Psi_1 = r\sin\frac{2\pi}{\lambda}(x - vt) = 2\sin\frac{2\pi}{12.5}(x - 25t)$$

and $$\Psi_1 = r\sin\frac{2\pi}{\lambda}(x + vt) = 2\sin\frac{2\pi}{12.5}(x + 25t).$$

1.13 Wave Packet

The concept of a wave packet is purely quantum mechanical. A pure sine wave extends from $-\infty$ to $+\infty$. That means it is completely unlocalized. A classical particle is